Operation Komorebi

Large Signal Model

Foundations of realization, refinement and predictive closure

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Exact representation. Observable learning.
Certified probability domains.
Typed defects and controlled approximation.

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This supplement develops the mathematical foundation of the Projector-Resolved Large Algebraic Signal Model (PR-LASM). It accompanies software release 2.3.0 without changing that release's implementation or historical verification records. Its subject is the structure of a signal model, not performance on a task suite.

The central question is not only whether a signal can be reconstructed. It is whether a chosen description retains enough information for the transformations and inferences performed on it. A representation can be exact as an archive and insufficient as a predictive state. A predictive state can be sufficient for a specified family of dynamics without reconstructing the entire signal.

The core organizing principle is therefore:

> Preserve the observed signal. Declare the representation used for reasoning. Express transformations in that representation. Identify the obstruction when transformation, compression or inference fails to commute. Refine the representation or retain memory when the obstruction matters.

The finite atlas, packet and best-basis results below consolidate the historical PR-LASM capstone [K1]. The Bayesian coarsening result is brought forward from Paper VI [K2]. The predictive-closure constructions connect these foundations to invariant-subspace and projection-memory theory [R2, R3]. Their role here is a self-contained foundation for the model, not a claim that these classical ingredients were first discovered by Komorebi.

The numbered results have proofs in this document. Exact rational and symbolic examples accompany them. They have not been newly formalized or checked by a Lean kernel in this supplement.

Edition 0.4.0 contains Propositions 1–49. Part I develops realization and closure. Part II begins at Section 15 with observable learning and stability. Part III begins at Section 27 with controlled stochastic realization. Part IV begins at Section 41 and adds finite probability-domain certificates, robustness, full-law error bounds and a nonpolyhedral boundary with an explicit approximation. The production software remains at version 2.3.0 and is not changed by this theory edition.

Sections 14, 26 and 40 retain the end-of-block checkpoints from preceding editions. Their stated open directions are historical waypoints. Part IV and Section 55 give the current boundary. The unchanged 0.3.0 archive is included for source lineage, and it contains the earlier archives. General proofs appear in the text, while computational records check explicit finite cases.

1. Three objects that must remain distinct

An **observation** is an element x of a declared space X. Examples include a finite real signal, a complex field on a finite grid, or an array of categorical labels. Its type includes shape, scalar field or palette, and boundary convention. A zero-extended finite sequence and a periodic sequence are different objects even when their displayed samples agree.

An **archive** is a lossless representation of the accepted observation. In a mathematical model this means an injective analysis map with an explicit left inverse. In an implementation, conversion and floating-point error must be accounted for separately. An archive of a measured signal does not recover physical variables that were never observed.

A **predictive state** is a summary $z = \gamma(x)$, or more generally a summary of an observation history. It is sufficient only relative to a specified family of future quantities, actions and dynamics. There is no implication from exact storage of x to sufficiency of $\gamma(x)$.

The historical model's macro, meso and micro vectors are nested projections of a feature vector. They are not, merely by being nested, state variables on which the signal dynamics close. The distinction is retained in the present theory.

The term **large** denotes the structured hypothesis family: representations, resolutions, typed operators, compositions, memory mechanisms and their parameters. It is not a theorem about capacity, sample complexity or scale. Such a theorem needs a declared grammar, a complexity measure and a learning protocol.

2. Exact realization and atlas consistency

2.1 A left inverse is the reconstruction contract

Let X and C_a be finite-dimensional Hilbert spaces. A linear atlas consists of analysis and synthesis maps

$$A_a : X \longrightarrow C_a, \qquad S_a : C_a \longrightarrow X, \qquad S_a A_a = I_X.$$

The coefficients representing actual signals form $\mathcal{C}_a = \operatorname{im} A_a$. This distinction matters for redundant representations, where $\mathcal{C}_a$ is a proper subspace of the ambient coefficient space.

Proposition 1. Exact realization and consistency. The analysis map is injective, $\Pi_a = A_a S_a$ is an idempotent with image $\mathcal{C}_a$, and $\Pi_a c = c$ exactly when c is a realizable coefficient vector. Moreover,

$$\|A_a x\| \geq \frac{\|x\|}{\|S_a\|}, \qquad \|S_a(c+e) - S_a c\| \leq \|S_a\|\,\|e\|.$$

The first inequality is understood for nonzero X. No assertion of orthogonality follows from a left inverse alone.

Proof. If $A_a x = 0$, then $x = S_a A_a x = 0$. Also $\Pi_a^2 = A_a(S_a A_a)S_a = \Pi_a$. Its image is contained in $\mathcal{C}_a$ and it fixes every $A_a x$, so its image is exactly $\mathcal{C}_a$. Both norm inequalities follow from the operator-norm inequality applied to S_a. □

For a unitary atlas U_a, the synthesis is U_a^* and the condition number is one. For a learned invertible atlas, exact invertibility is not enough: the synthesis norm controls amplification of coefficient errors. An arbitrarily ill-conditioned exact transform is not a stable representation.

2.2 Several exact atlases do not create new observations

For unitary U_a and positive weights w_a, put

$$\mathcal{A}x = (w_a U_a x)_a, \qquad \mathcal{S}(c_a)_a = \frac{1}{c}\sum_a w_a U_a^* c_a, \qquad c = \sum_a w_a^2.$$

Then $\mathcal{S}\mathcal{A} = I$ and $\|\mathcal{A}x\|^2 = c\|x\|^2$, as proved in [K1]. Here $\mathcal{A}\mathcal{S} = c^{-1}\mathcal{A}\mathcal{A}^*$ is the orthogonal projection onto the consistent coefficient subspace.

Each complete atlas contains the same observed information because it has a left inverse. Multiple atlases offer different sparsity, locality, conditioning and operator descriptions. They do not provide independent observations of the world. In particular, multiplying likelihoods from deterministic recodings as though they were independent measurements generally counts the same evidence more than once.

Categorical one-hot encoding is a separately typed injective map of sets. It is not a unitary map on the scalar-valued signal space. Its exact decoder is valid on one-hot arrays, not on arbitrary coefficient vectors. The linear tight-frame theorem does not apply to it without a separately specified linear embedding and inner product. This separation is already explicit in the capstone's categorical section.

3. Coordinate transport and one underlying operator

On realizable coefficients, define transport from atlas a to atlas b by

$$T_{ba} = A_b S_a : \mathcal{C}_a \longrightarrow \mathcal{C}_b.$$

For an endomorphism $F : X \to X$, define its coordinate realization by $F_a = A_a F S_a$ on $\mathcal{C}_a$. In this paragraph F may be nonlinear. For a partial map, every formula is restricted to the transported admissible domain. Maps between different signal types use separate input and output atlases.

Proposition 2. Coherent transport and operator realization. On the consistent coefficient spaces,

$$T_{aa} = I, \qquad T_{cb}T_{ba} = T_{ca}, \qquad F_b T_{ba} = T_{ba} F_a.$$

Conversely, suppose maps $G_a : \mathcal{C}_a \to \mathcal{C}_a$ satisfy $G_b T_{ba} = T_{ba} G_a$ for every pair. Then there is a unique map $F : X \to X$ with $G_a = A_a F S_a$ for every a.

Proof. The transport identities follow by inserting $S_b A_b = I$. The operator identity follows because either path is $A_b F S_a$. For the converse, choose a reference atlas r and put $F = S_r G_r A_r$. Compatibility gives $G_a A_a = A_a F$. Every consistent coefficient is $A_a x$, which proves the asserted realization. Applying S_a proves uniqueness. □

An independently fitted family of atlas predictors need not satisfy this proposition. Its compatibility residual is a well-defined quantity,

$$d_{ba}(x) = G_b(A_b x) - T_{ba} G_a(A_a x).$$

After synthesis it becomes an object-space discrepancy:

$$S_b d_{ba}(x) = S_b G_b A_b x - S_a G_a A_a x.$$

Thus $\|S_b d_{ba}(x)\| \le \|S_b\| \, \|d_{ba}(x)\|$. Zero discrepancy establishes that the chart descriptions agree, not that their common prediction is correct. All charts can agree on the same wrong operator.

Redundant coefficient outputs must also be tested for consistency. Synthesizing an arbitrary coefficient vector always produces some signal, but reanalysis returns $\Pi_a c$, not necessarily the original c. Reconstruction and coefficient consistency are different checks.

4. Packets, refinement and transfer blocks

4.1 Packets belong to a declared resolution

Within one orthogonal atlas, let E_i be pairwise orthogonal coordinate selectors with $\sum_i E_i = I$. Pull them back by a unitary map:

$$P_i = U^* E_i U, \qquad P_i P_j = \delta_{ij} P_i, \qquad P_i^* = P_i, \qquad \sum_i P_i = I_X.$$

A packet is the component $P_i x$ together with its projector, domain, atlas and resolution. A bare coefficient block without this information is not a fully specified packet.

Projectors from different atlases need not commute and must not be treated as one larger mutually orthogonal family. Weighted frame packets can sum to the identity while failing to be idempotent. These are different structures.

Proposition 3. Complete refinement preserves reconstruction. Suppose a finite rooted tree has root projector I, and every internal projector is the sum of pairwise orthogonal child projectors. Every complete antichain Λ, meeting each root-to-leaf path exactly once, satisfies

$$\sum_{v \in \Lambda} P_v = I, \qquad x = \sum_{v \in \Lambda} P_v x, \qquad \|x\|^2 = \sum_{v \in \Lambda} \|P_v x\|^2.$$

Proof. Replace an internal node by the sum of its children. Repeated replacement reaches any complete antichain and preserves the total projector. Different selected nodes have disjoint descendant branches, hence orthogonal ranges. The norm identity is Pythagoras. □

This is a conservation law for information under complete refinement. Deleting a child, storing only its energy, or imposing a coefficient threshold changes the contract. For a retained subset J of an orthogonal decomposition, the lost signal is exactly $\sum_{i \notin J} P_i x$, and its squared norm is the sum of the omitted packet energies.

The finite best-basis recursion

$$V(v) = \min\left(c(v), \sum_{u \in \mathrm{ch}(v)} V(u)\right)$$

minimizes an additive cost over the declared finite tree [K1, R1]. The recurrence itself does not make a nonadditive predictive objective additive. If a prediction cost couples two leaves, a scalar cost at each node is generally insufficient for global optimization. Packet construction and cost evaluation must also be included in an implementation's complexity account.

4.2 Operations, not just signals, decompose

Let $L : X \to Y$ be linear, with orthogonal input projectors P_i and output projectors Q_j. Its transfer blocks are $L_{ji} = Q_j L P_i$.

Proposition 4. Transfer reconstruction and refinement. One has

$$L = \sum_{j,i} Q_j L P_i.$$

If $P_i = \sum_r P_{i,r}$ and $Q_j = \sum_s Q_{j,s}$ are complete refinements, then

$$Q_j L P_i = \sum_{s,r} Q_{j,s} L P_{i,r}.$$

For another linear map $M : Y \to Z$ and output projectors R_k,

$$R_k M L P_i = \sum_j (R_k M Q_j)(Q_j L P_i).$$

Proof. Insert the appropriate identity decomposition on the input, output or intermediate space. Use $Q_j^2 = Q_j$ in the composition formula. □

This is the algebraic reason to make transfer blocks first-class objects. The same operation can be expressed at several resolutions, while the coarse and fine expressions remain linked by exact identities. Approximations acquire explicit omitted-block terms rather than an unnamed representation error.

4.3 Nonlinear operations need interactions

For a nonlinear map, $F(\sum_i P_i x)$ generally differs from $\sum_i F(P_i x)$. Applying a linear transfer formula to such a map silently loses interactions.

Proposition 5. Polynomial packet interactions. Let

$$F(x) = b + \sum_{m=1}^{p} B_m(x, \ldots, x),$$

where each $B_m : X^m \to Y$ is multilinear. Then

$$Q_j F(x) = Q_j b + \sum_{m=1}^{p} \sum_{i_1, \ldots, i_m} Q_j B_m(P_{i_1} x, \ldots, P_{i_m} x).$$

Proof. Substitute $x = \sum_i P_i x$ in every argument and distribute by multilinearity. □

For example, $(u+v)^2 = u^2 + 2uv + v^2$. A grammar containing only independent packet squares misses $2uv$. For nonpolynomial maps, one needs direct nonlinear semantics or a local expansion with a stated remainder bound. The word algebraic does not remove that obligation.

5. Predictive closure is the central contract

Let $\gamma : X \to Z = \gamma(X)$ be a summary and let $F_u : X \to X$ describe a deterministic transition for action u. The reduced description closes when there is $\bar{F}_u : Z \to Z$ such that

$$\gamma F_u = \bar{F}_u \gamma.$$

This is an equality of functions on the declared domain, not a fit statistic.

Proposition 6. Exact quotient criterion. Such a reduced transition exists if and only if

$$\gamma(x) = \gamma(x') \implies \gamma(F_u(x)) = \gamma(F_u(x'))$$

for every admissible x, x' and every action. When it exists it is unique on $\gamma(X)$.

Proof. Necessity follows by applying $\bar{F}_u$ to equal summaries. For sufficiency, define $\bar{F}_u(z) = \gamma(F_u(x))$ for any representative with $\gamma(x) = z$. The stated condition makes this well-defined and also proves uniqueness. □

This criterion also applies when X is a latent physical state and γ is an observation map. Two physically different states can produce identical images and different futures. Exact image reconstruction does not resolve that hidden-state ambiguity. A history, a belief state or additional observations may be needed.

For a stochastic kernel, the corresponding condition is equality of the distributions of the next summary for states with the same current summary. Section 7 makes this finite criterion explicit.

5.1 The obstruction in the linear case

Suppose X is a Hilbert space, P is an orthogonal projection, $Q = I - P$ and $x_{t+1} = Lx_t$. The retained state is $z_t = Px_t$. Then

$$z_{t+1} = PLPz_t + PLQx_t.$$

Proposition 7. Closure defect and compositional defect. The retained state closes for every initial signal if and only if $PLQ = 0$. If it does not close, the error in using only PLP is

$$e_t = PLQx_t, \qquad \|e_t\| \le \|PLQ\| \, \|Qx_t\|.$$

For two linear operators L and M,

$$\boxed{PLMP - (PLP)(PMP) = PLQMP.}$$

Consequently, compressing two operations separately can discard a path that leaves the retained subspace and later returns.

Proof. If $PLQ = 0$, the displayed recurrence depends only on z_t. Conversely, apply Proposition 6 to x and Px, which have the same retained state, to obtain $PLQx = 0$ for every x. The norm bound is submultiplicativity. The compositional identity follows by inserting $I = P + Q$ between L and M. □

The relevant quantity is not omitted energy alone. A very small omitted packet can have a large effect if its transfer into retained packets is strongly amplified. Conversely, an energetic packet can be irrelevant to a specified prediction if every corresponding return block is zero.

A residual is therefore typed. Reconstruction residual, coefficient-consistency residual, predictive residual, closure residual and statistical uncertainty should not be collapsed into one scalar without a specified objective.

6. Constructing the smallest closed linear refinement

There is a constructive finite solution when the dynamics are a known finite family of linear maps L_u on a d-dimensional space. It identifies which additional observables are required, rather than merely requesting a larger state vector.

Let the initial summary be C_0x. Regard its rows as linear observables. Starting from $W_0 = \mathrm{row}(C_0)$, define

$$W_{n+1} = W_n + \sum_u W_n L_u, \qquad W_n L_u = \{wL_u : w \in W_n\}.$$

Proposition 8. Minimal linear predictive completion. The sequence stabilizes after at most $d - \dim W_0$ strict enlargements. Its limit W_* is the smallest row space containing W_0 and invariant under right multiplication by every L_u. If R is any full-row-rank matrix whose rows form a basis of W_*, then

$$C_0 = HR, \qquad RL_u = B_u R$$

for suitable matrices H and B_u. Therefore $z = Rx$ retains the original summary and has exact reduced dynamics $z_{t+1} = B_u z_t$. Every linear summary with those two properties has row space containing W_*.

Proof. The spaces are nested and have dimension at most d. Equality at one step means invariance under every L_u, so it is a fixed point. Containment in any invariant row space containing W_0 follows by induction. The matrix identities express inclusion of the relevant rows in the basis R. Conversely, if $C_0 = HR$ and $RL_u = B_u R$, the row space of R is invariant and contains W_0, so it contains every W_n. □

For one transition matrix this reduces to the span of the rows of

$$C_0,\ C_0L,\ \ldots,\ C_0L^{d-1}.$$

Its annihilator is the largest L-invariant subspace inside $\ker C_0$, the standard unobservable-subspace construction [R2]. With several actions, observables pulled back through action words are required, but the linear span still stabilizes in finite dimension.

This gives a precise role for obstruction-driven refinement: add observable directions created by the dynamics until the chosen prediction variables close. It does not guarantee that the completion is small. In some systems $W_* = X^*$ and exact memoryless closure needs the full state.

The theorem assumes the operators and relevant state are available. Applying it to estimated operators constructs closure for those estimates. It is not a proof that the real system has the same closure. If only C_0x is measured and the full x is unavailable, the augmented state Rx is not automatically accessible.

7. Joint closure of prediction and Bayesian inference

The finite categorical case admits an equally explicit construction that joins dynamics with Paper VI's inference refinement. This is useful because a state can be closed for prediction but too coarse for a new observation, or sufficient for an update but not for the next transition.

Let Ω be a finite state set. Let $K_u(i,j)$ be known row-stochastic transition matrices. Let $\ell_y(i) > 0$ be a finite family of observation likelihoods. A partition π has aggregation matrix

$$C_\pi(a,i) = \mathbf{1}_{i \in B_a}.$$

Distributions are column vectors. Prediction is $p \mapsto K_u^\top p$, and a Bayesian update is

$$\mathcal{B}_{\ell_y}(p) = \frac{p \odot \ell_y}{\ell_y^\top p}.$$

The coarse state must commute with both operations:

$$C_\pi K_u^\top = \bar{K}_u^\top C_\pi, \qquad C_\pi \mathcal{B}_{\ell_y}(p) = \mathcal{B}_{\bar{\ell}_y}(C_\pi p).$$

Fix a mandatory initial partition π_0 with at least two blocks. This preserves distinctions that the intended application requires and excludes the trivial one-state quotient.

Proposition 9. Coarsest joint finite closure. First split π_0 by the likelihood vectors $(\ell_y(i))_y$. Then repeatedly split each current block by the signatures

$$\left(\sum_{j \in B} K_u(i,j) \right)_{u,B},$$

where B ranges over the current partition. The process terminates after at most $|\Omega| - |\pi_{\text{init}}|$ strict refinement rounds. The final partition is the unique coarsest refinement of π_0 on which all declared predictions and Bayesian updates close for every prior.

Proof. At a stable partition, states in a common block have the same transition probability into every coarse block. Those probabilities define $\bar{K}_u$ and give the first commutation identity, by checking it on point-mass priors. Likelihood constancy on each block defines $\bar{\ell}_y$ and gives the second identity by direct summation.

For minimality, every admissible partition refining π_0 has at least two blocks. By Proposition 12 below, exact updating for all priors requires each fine likelihood to be constant within each block. Thus it refines the first likelihood split. Suppose it refines a current partition. Each current block is a union of its blocks. Exact prediction makes the probability of reaching each such union constant within any admissible block, so it also refines the next split. Induction proves it refines the final partition. Each strict refinement increases the number of blocks, which proves termination. □

The transition condition is the finite lumpability criterion [R4], expressed directly as a commutation identity. The joint construction adds the explicitly declared likelihood distinctions. Its output is a quotient of a known finite model, not an automatically learned model of an unknown world.

The accompanying rational example starts with six states. Likelihoods create three blocks, transition distinctions force two further rounds of splitting, and the final model has five states. Two original states remain genuinely equivalent for every action and observation in the declared family. The example verifies both commuting identities exactly.

8. Memory is an exact alternative to adding state

Refinement and memory are two ways to represent the influence of unresolved components. This can be derived rather than merely proposed as an architectural option.

For fixed linear dynamics and an orthogonal split, write

$$z_t = Px_t, \qquad q_t = Qx_t,$$

and define the restricted blocks

$$A = PLP, \quad B = PLQ, \quad C = QLP, \quad D = QLQ.$$

Thus $z_{t+1} = Az_t + Bq_t$ and $q_{t+1} = Cz_t + Dq_t$.

Proposition 10. Exact projected memory and a conditional tail bound. For every $t \geq 0$,

$$\boxed{z_{t+1} = Az_t + \sum_{k=0}^{t-1} BD^k Cz_{t-1-k} + BD^t q_0.}$$

If $\|D\| \leq \rho < 1$ and $\|z_s\| \leq Z$ over the relevant history, omitting terms with $k \geq H$ changes the one-step right-hand side by at most

$$\|B\|\,\|C\| Z \frac{\rho^H}{1-\rho}.$$

Dropping the initial unresolved-state term adds at most $\|B\|\rho^t\|q_0\|$.

Proof. Iteration of the unresolved recurrence gives

$$q_t = D^t q_0 + \sum_{s=0}^{t-1} D^{t-1-s} Cz_s.$$

Substitute this into the retained recurrence and change indices. Bound the omitted terms by the geometric series. □

The memory kernel is BD^kC. It records a path from retained variables into unresolved variables, evolution there, and return. The first kernel BC is precisely the return path appearing in the two-step composition defect. The term BD^tq_0 is deterministic when q_0 is specified. It becomes uncertain only relative to incomplete information or a probability model for q_0.

This is the elementary linear instance of projection-memory reduction, associated with Mori-Zwanzig theory [R3]. A finite hand-chosen power-law or exponential memory feature is an approximation class for such effects, not a general derivation of the exact kernel.

The tail bound compares right-hand sides using the same resolved history. Replacing that history by recursively generated predictions introduces additional rollout error. Stability of that recursion needs its own bound. Also, a spectral-radius condition alone does not imply $\|D\| \le \rho$ in the norm already being used, especially for nonnormal operators.

9. Statistical refinement needs its own theorem

Nested feature vectors are algebraically nested when the smaller vector can be recovered from the larger by a fixed map. Predictive risk is a separate matter.

Proposition 11. Ideal statistical benefit of refinement. Let a target Y take values in a finite-dimensional Hilbert space with $\mathbb{E}\|Y\|^2 < \infty$. Let $\mathcal{F} \subseteq \mathcal{G}$ be information sets and define $m_{\mathcal{F}} = \mathbb{E}[Y \mid \mathcal{F}]$ and $m_{\mathcal{G}} = \mathbb{E}[Y \mid \mathcal{G}]$. Then

$$\mathbb{E}\|Y - m_{\mathcal{F}}\|^2 = \mathbb{E}\|Y - m_{\mathcal{G}}\|^2 + \mathbb{E}\|m_{\mathcal{G}} - m_{\mathcal{F}}\|^2.$$

Proof. Expand $Y - m_{\mathcal{F}} = (Y - m_{\mathcal{G}}) + (m_{\mathcal{G}} - m_{\mathcal{F}})$. The cross term has zero conditional expectation given $\mathcal{G}$. □

Thus finer information cannot worsen the optimal squared-error predictor under a fixed distribution. It can worsen an estimated predictor with finite data, restricted function classes, changed regularization or shifted distributions. The theorem supplies the ideal target for refinement, not a guarantee for every fitting procedure.

An exact archive makes later reanalysis possible without reconstructing discarded observations from a lossy state. It does not eliminate estimation error or distribution shift.

10. Bayesian coarsening and the information-loss packet

This section imports a specific existing Komorebi foundation rather than inventing another inference heuristic. Let p_i be a prior on a finite fine state set and let B_a be coarse blocks. Define

$$P_a = \sum_{i \in B_a} p_i, \qquad r_a(p) = \frac{\sum_{i \in B_a} p_i \ell_i}{P_a} \quad (P_a > 0).$$

On zero-mass blocks the value of r_a is immaterial. Updating finely and then aggregating is exactly updating P with the effective likelihood $r(p)$. The effective likelihood generally depends on the fine prior inside each coarse block, information that P alone does not retain.

Proposition 12. Bayesian closure and its exact defect. Suppose there are at least two coarse blocks and the fine and declared coarse likelihoods are strictly positive. Updating and coarsening commute for every prior if and only if

$$\ell_i = c\bar{\ell}_a \quad \text{for every } i \in B_a$$

with one common $c > 0$. For a fixed prior, put $\delta = r(p) - \bar{\ell}$, $\bar{b} = \mathcal{B}_{\bar{\ell}}(P)$ and $Z_r = r(p)^\top P$. Then

$$\boxed{C_\pi \mathcal{B}_\ell(p) - \bar{b} = \frac{1}{Z_r}\left(\operatorname{diag} P - \bar{b}P^\top\right)\delta.}$$

The fixed-prior identity also holds for nonnegative likelihoods when both normalizing evidences are positive.

Proof. Constancy up to one global factor gives commutation by summing the fine posterior. Conversely, compare priors supported on one state in each of two distinct coarse blocks. Equality of coarse posterior odds forces the ratio of those states' likelihoods to equal the ratio of the coarse likelihoods. Varying the representatives proves constancy within each block and the common factor across blocks. For the defect, use $Z_r = \bar{\ell}^\top P + P^\top \delta$ and subtract the two normalized vectors. □

With only one coarse block, both posteriors aggregate to one regardless of the likelihood, so the necessity criterion is false there. This boundary is explicitly retained from the corrected Paper VI statement [K2].

The matrix in the defect formula annihilates $\bar{\ell}$, recording the irrelevant common-scale direction. It does not make every likelihood discrepancy irrelevant. The construction in Proposition 9 refines until both the likelihood and transition defects vanish for a specified finite family.

10.1 Parameter inference is not program truth

Within one declared regression model, a normal-inverse-gamma prior gives a standard conditional Bayesian parameter posterior. The software's program weights instead use penalized held-out predictive scores and then may refit coefficients on all available training records. These are finite-candidate predictive weights. Without a corresponding generative derivation, they should not be conflated with the marginal-likelihood posterior over programs under one fixed prior.

Every candidate must be scored on the same target event and base measure. Densities on differently sized coordinate spaces are not directly comparable. An invertible nonlinear coordinate change requires the appropriate Jacobian. A redundant deterministic embedding induces a measure on its consistent subspace, not an independent full-dimensional density on all coefficient entries.

The law of total variance separates within-candidate uncertainty from disagreement of candidate means. It does not supply a probability that the truth lies outside the grammar. An explicit outside-grammar component needs its own prior and predictive distribution. Assigning an arbitrary leftover mass does not solve open-world inference.

11. What a typed operator language must guarantee

A program has a source type, target type, admissible domain, evaluation rule and declared approximation status. A nonlinear partial operation is not a total matrix merely because its inputs are stored in a vector. The domain of a composite is

$$\operatorname{dom}(g \circ f) = \{x \in \operatorname{dom} f : f(x) \in \operatorname{dom} g\}.$$

Composition must preserve those types and domains. Coordinate changes must preserve the represented operation as in Proposition 2. Packet refinement must preserve exact transfer semantics as in Proposition

4, or carry an explicit approximation residual. Nonlinear interactions must retain their cross-packet terms as in Proposition 5.

A grammar can be closed under a specified set of constructors without containing every possible physical law. It can also be expressive while computationally intractable. Universal discovery, practical search and semantic closure are distinct claims.

For a known operator, Proposition 8 can build its minimal linear summary. For an unknown operator, learning must address identifiability: which different candidates make identical predictions under the available observations and actions? More coordinate systems alone do not distinguish candidates that agree on the entire observed signal. Additional interventions or data may be needed.

The foundation should therefore treat a learned hypothesis as an operator with an evidence record, not just a matrix with a high normalized weight. Its record includes fitting assumptions, admissible inputs, compared alternatives, unresolved equivalences and the representation in which its residual is measured.

12. A bounded infinite-resolution bridge

Finite reconstruction identities do not alone prove convergence of an unbounded or infinite-dimensional model. A useful bounded extension can nevertheless be established directly.

Proposition 13. Strong fixed-horizon refinement limit. Let X be a Hilbert space, $L : X \to X$ bounded, and P_n orthogonal projections converging strongly to I. Put $L_n = P_n L P_n$. For every fixed $x \in X$ and every fixed nonnegative integer k,

$$L_n x \longrightarrow Lx, \qquad L_n^k x \longrightarrow L^k x.$$

Proof. Since $\|P_n\| \le 1$,

$$\|(L_n - L)x\| \le \|L\|\,\|(P_n - I)x\| + \|(P_n - I)Lx\| \longrightarrow 0.$$

Also $\|L_n\| \le \|L\|$. Induct on k, using

$$(L_n^{k+1} - L^{k+1})x = L_n(L_n^k - L^k)x + (L_n - L)L^k x.$$

The first term tends to zero by the uniform norm bound and induction, and the second by strong convergence on the fixed vector $L^k x$. □

If estimated operators $\widehat{L}_n$ are uniformly bounded and $(\widehat{L}_n - L_n)x \to 0$ for every fixed x, the same argument gives fixed-horizon convergence for $\widehat{L}_n$. These are explicit assumptions, not conclusions of the finite learning procedure.

This result is strong convergence, not operator-norm convergence and not a bound uniform in time. On ℓ^2, finite coordinate projections converge strongly to I while $\|I - P_n\| = 1$ for every n. Unbounded operators require domain control. Adaptive data-dependent partitions and fitted coefficients require additional consistency arguments. These remain separate mathematical tasks.

13. A complete small signal example

Consider the rational linear signal dynamics

$$x_{t+1} = Lx_t, \qquad L = \begin{pmatrix} \frac{1}{2} & 1 \\ \frac{1}{4} & \frac{1}{3} \end{pmatrix}, \qquad x_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

The spatial archive stores x_t. A second atlas uses the orthogonal Haar matrix

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Its coefficients $c_t = Ux_t$ reconstruct as $x_t = U^* c_t$. The same dynamics in that atlas are ULU^*, not a separate physical law. Thus coordinate transport preserves the full dynamics.

Now retain only the first spatial coordinate. With $P = \mathrm{diag}(1, 0)$, the retained scalar z_t and discarded scalar q_t obey

$$z_{t+1} = \tfrac{1}{2} z_t + q_t, \qquad q_{t+1} = \tfrac{1}{4} z_t + \tfrac{1}{3} q_t.$$

The exact first-coordinate sequence begins

$$z_0 = 1, \qquad z_1 = \tfrac{1}{2}, \qquad z_2 = \tfrac{1}{2}, \qquad z_3 = \tfrac{11}{24}.$$

The compressed one-step matrix predicts $z_2 = 1/4$ instead of $1/2$. The lost return path contributes exactly

$$BC = 1 \cdot \tfrac{1}{4} = \tfrac{1}{4}.$$

There are two exact repairs. One is state refinement: start with $C_0 = (1, 0)$ and add $C_0 L = (1/2, 1)$. The resulting row space is the full two-dimensional space, so it closes. The other is memory:

$$z_{t+1} = \tfrac{1}{2} z_t + \tfrac{1}{4} \sum_{k=0}^{t-1} \left(\tfrac{1}{3}\right)^k z_{t-1-k},$$

since $q_0 = 0$. This is the exact same system expressed without retaining q_t as a separate current-state variable. At $t = 2$, it yields

$$z_3 = \tfrac{1}{2} \cdot \tfrac{1}{2} + \tfrac{1}{4} \cdot \tfrac{1}{2} + \tfrac{1}{12} \cdot 1 = \tfrac{11}{24}.$$

To see a nontrivial reduced closure, adjoin a third independent coordinate with multiplier $5/6$. The smallest closed refinement of the first coordinate then contains the first two coordinates, not the third. Exact prediction of the chosen summary does not require reconstructing every component of the enlarged signal.

Finally, low energy is not the same as low predictive importance. For

$$L' = \begin{pmatrix} 0 & 1000 \\ 0 & 0 \end{pmatrix}, \qquad x = (0, 1/1000)^\top,$$

omitting the second coordinate loses norm $1/1000$, but changes the next first coordinate by exactly one. The correct certificate is the product of omitted magnitude and transfer amplification.

14. First-block checkpoint and release organization

The principal theoretical object is now a signal realization with explicit compression maps and commuting requirements. Exact archives, packets, operator descriptions and inference are connected by those requirements rather than simply listed as neighboring modules.

The inherited core supplies exact finite atlases, complete packet synthesis, additive best-basis selection and a finite typed predictive grammar [K1, K3]. Paper VI supplies the exact inference-coarsening defect and its hypotheses [K2]. This supplement develops a connected treatment of atlas coherence, predictive quotients, compositional defects, minimal closure, joint finite stochastic closure, projection memory and bounded refinement limits.

The additions are mathematical statements and reference calculations. They do not silently make the 2.3.0 runtime enforce atlas coherence, compute minimal predictive closures or discover the true transition law. An implementation map is supplied separately. The original source and historical manuscript remain unchanged.

A foundations-centered release should have one primary theory volume organized by realization, packets, transport, closure, memory and inference. The mathematical examples and proof ledger belong alongside it. Software documentation should identify which definitions and laws each component implements. Empirical material can remain supplementary without being the criterion for mathematical completeness.

The first edition identified the next substantive gaps as stable learned atlases, identification from incomplete observations, nonlinear interaction truncation, coherent program weighting under grammar growth, and adaptive refinement with learned operators. Part II now addresses explicit finite or bounded sectors of these questions. Its final section distinguishes those closed statements from the general problems that remain.

The finite theory can be stated decisively: exact realization preserves observations, complete refinement preserves decomposition, coherent transport preserves the meaning of operators, and predictive or inferential closure holds precisely under the corresponding compatibility conditions. Where closure fails, the theory supplies a typed defect and, in the finite known-model cases above, a constructive repair.

Part II. Learning, stability and coherent expansion

The first stage constructed predictive closures for declared operators. The second stage asks how those constructions relate to what a learner can actually observe. A closure that depends on an unavailable hidden coordinate is a mathematical description, not yet an accessible state. A fitted recurrence can be accessible but wrong outside the span represented by its data. A one-step approximation can be accurate and still fail under repeated composition.

The results below connect these distinctions. They give an exact learning result for a finite linear observable sector, deterministic perturbation bounds, a propagation law for typed defects, a sufficient stability condition for truncated memory, controlled nonlinear interaction bounds, coherent atlas reconciliation and posterior accounting for an expanding grammar. These are finite or explicitly bounded results. They do not assume that every unknown process has a short linear state.

The observable-state perspective has classical antecedents in predictive state representations [R5] and observable-operator learning [R7]. Finite invariant observable spaces also underlie Koopman reductions [R6]. The present development uses elementary statements proved below and identifies their specific place in Komorebi's realization and obstruction calculus. It does not import those papers' sample-complexity guarantees into this model.

15. From hidden state to an accessible predictive state

Let a finite-dimensional autonomous linear process have hidden state $x_t \in \mathbb{R}^d$ and observed output $y_t \in \mathbb{R}^p$:

$$x_{t+1} = Lx_t, \qquad y_t = Cx_t.$$

The observation map need not be injective. Define the length-h observation map

$$O_h = \begin{pmatrix} C \\ CL \\ \vdots \\ CL^{h-1} \end{pmatrix}, \qquad V_h = \operatorname{im} O_h.$$

The coordinates of $O_h x_s$ are future observations relative to time s. They are not available at time s merely because this matrix is written down. At time $t \geq h-1$, however, the same vector is the observed history

$$z_t = \begin{pmatrix} y_{t-h+1} \\ \vdots \\ y_t \end{pmatrix} = O_h x_{t-h+1}.$$

Thus a forward observation map becomes a causal state after a declared warm-up interval. The time index is part of the contract.

Proposition 14. Exact causal delay closure. The following statements are equivalent:

1. There is a linear map $D_h : V_h \to V_h$ satisfying $O_h L = D_h O_h$.
2. $\ker O_h \subseteq \ker(O_h L)$.
3. $\operatorname{rank} O_{h+1} = \operatorname{rank} O_h$.

When these statements hold, D_h is unique on V_h. The history state obeys $z_{t+1} = D_h z_t$, and its last output block is y_t. Its iterates therefore determine every subsequent observed output. If d is an upper bound on the hidden dimension, $h = d$ is sufficient for closure, though a smaller h may suffice.

Proof. The equivalence of the first two statements is the linear factorization criterion: define $D_h(O_h x) = O_h L x$ and require independence from the representative x. The rows of $O_h L$ are $CL, \ldots, CL^h$. All except the final block already occur in O_h. The kernel condition is therefore equivalent to CL^h being in the row space of O_h, which is exactly the rank equality. It also implies $O_h Lx \in V_h$ because this vector is O_h applied to Lx. Uniqueness follows because every element of V_h has a representative. The history recurrence follows directly. Cayley-Hamilton expresses L^d as a linear combination of lower powers, so CL^d is in the row space of O_d. □

This does not recover hidden variables in $\ker O_h$. It identifies the quotient needed for the declared outputs. An independent hidden component can remain unobservable forever without preventing exact output prediction.

The rank equality here is a property of the *true observation maps*. A plateau in a finite empirical data matrix is not the same statement. It may reflect unexcited directions rather than a closed observation space. Section 17 makes the difference exact.

For several actions L_u, a fixed delay vector does not automatically have one autonomous transition matrix. Products of the actual action matrices determine its observation map, which can change with the action history. One needs an action-aware state, a common closed observable family, or a separately justified belief-state construction. Proposition 14 is not silently generalized to arbitrary controlled or stochastic dynamics.

16. What finite data identify

Suppose candidate states and their successors have been recorded as columns of matrices $Z_-, Z_+ \in \mathbb{R}^{m\times N}$. Their construction may use output histories. No hidden-state coordinates need be supplied to the fitting procedure.

Write $W = \operatorname{im} Z_-$. Exact interpolation of these columns is a question about a linear map on W, not yet about every physically realizable state.

Proposition 15. Data-span identification and its boundary. A linear map B on the ambient space with $BZ_- = Z_+$ exists if and only if

$$\ker Z_- \subseteq \ker Z_+.$$

Its restriction to W is unique. It is an endomorphism of W if and only if $\operatorname{im} Z_+ \subseteq W$. A particular ambient extension is

$$B_0 = Z_+ Z_-^\dagger,$$

where $\dagger$ denotes the Moore-Penrose inverse. Every other fitted extension agrees with B_0 on W. If W is a proper subspace of the true realizable state space V, the data alone do not identify the operator on $V \setminus W$.

Proof. Necessity follows by applying B to every linear dependence among the input columns. Under the kernel condition, define $B(Z_-a) = Z_+a$. Two choices of a differ by a kernel vector and give the same answer. This defines and uniquely determines the map on W. It maps W into W exactly under the image condition. The pseudoinverse formula gives that map because $a - Z_-^\dagger Z_- a \in \ker Z_-$. If $W \subsetneq V$, choose a nonzero linear functional φ on V vanishing on W and a nonzero $v \in V$. The operators B and $B + v\varphi$ agree on every observed column and differ elsewhere. □

The three checks are different. The kernel test checks consistency of a deterministic linear fit. The image test checks whether the observed span is closed under the fitted transitions. Equality $W = V$ is a *coverage condition*. Neither of the first two checks establishes it.

For a known fixed feature space of dimension r, coverage is witnessed by r independent input states. For an unknown larger feature space, the same rank only gives a lower bound on dimension. A model card must say which interpretation is being used.

If several action-specific operators act on a common declared r-dimensional state, each action's paired data must determine that operator on the required span. Pooling all actions into one high-rank matrix does not identify an action that has never been tried on some direction.

16.1 Perturbations have an explicit amplification factor

A noisy rank test is not an exact certificate. Even after the state space is fixed, fitting can be ill-conditioned. The following deterministic bound isolates that issue without introducing a probability model for noise.

Work in a fixed r-dimensional coordinate system, where $Z_- \in \mathbb{R}^{r\times N}$ has full row rank. Let

$$Z_+ = BZ_- + R, \qquad \widetilde{Z}_- = Z_- + E, \qquad \widetilde{Z}_+ = Z_+ + F.$$

The matrix R records deviations from the declared linear recurrence on these samples. The matrices E and F record input and successor perturbations. All norms in the next proposition are spectral norms.

Proposition 16. Fixed-space identification bound. Put $s = \sigma_{\min}(Z_-) > 0$. If $\|E\| < s$, then $\widetilde{Z}_-$ has full row rank, and

$$\widehat{B} = \widetilde{Z}_+ \widetilde{Z}_-^{\dagger}$$

satisfies the exact error identity and bound

$$\widehat{B} - B = (R + F - BE)\widetilde{Z}_-^{\dagger},$$
$$\|\widehat{B} - B\| \le \frac{\|R\| + \|F\| + \|B\|\,\|E\|}{s - \|E\|}.$$

Proof. For every unit $v \in \mathbb{R}^r$, $\|\widetilde{Z}_-^{\top} v\| \ge \|Z_-^{\top} v\| - \|E^{\top} v\| \ge s - \|E\|$. Thus the perturbed matrix has full row rank and its pseudoinverse has norm at most $(s - \|E\|)^{-1}$. Since $\widetilde{Z}_- \widetilde{Z}_-^{\dagger} = I_r$, subtract B from the fitted expression and collect terms. The bound follows by submultiplicativity and the triangle inequality. □

This is a bound for a fixed state space and a fixed coordinate basis. If the basis or its dimension is also estimated from noisy data, subspace-estimation error must be added. No such error is hidden inside R without a derivation. Correlated and adversarial perturbations are allowed by the displayed deterministic inequality, but there is no sample-complexity claim until their magnitudes are bounded under a sampling model.

The denominator explains why numerical rank alone is insufficient. Full rank can coexist with severe noise amplification. The reference calculations include examples attaining the $1/s$ dependence and a rank-collapse example at the excluded boundary $\|E\| = s$.

17. Why finite agreement is not unrestricted discovery

Proposition 17. Finite-prefix indistinguishability. For every integer $T \ge 0$, there are two finite-dimensional autonomous rational linear systems with identical scalar observations at times $0, \dots, T$ and different observations at time $T + 1$. Both transition matrices can be chosen nilpotent.

Proof. One system has scalar output identically zero. For the other, take dimension $T + 2$, initial state $x_0 = e_{T+2}$, observation row $C = e_1^{\top}$, and shift $Le_j = e_{j-1}$ for $j \ge 2$, $Le_1 = 0$. Then $CL^t x_0 = 0$ for $0 \le t \le T$ and $CL^{T+1} x_0 = 1$. The shift is rational and nilpotent. □

An invisible delayed component of this kind can also be adjoined to any fixed base model. The combined output agrees with the base output on the entire inspected prefix and then differs. This is not an

argument against learning. It identifies why a claimed guarantee must state a dimension bound, a model class, a coverage condition, an intervention protocol or a probabilistic assumption.

Even nilpotent operators have only zero eigenvalues, so stability language alone does not remove this ambiguity. A finite rank plateau cannot certify the absence of a delayed mode when arbitrarily large hidden dimensions are admitted.

18. A complete output-only identification example

Return to the rational system from Section 13, now with an additional independent hidden coordinate:

$$L = \begin{pmatrix} 1/2 & 1 & 0 \\ 1/4 & 1/3 & 0 \\ 0 & 0 & 5/6 \end{pmatrix}, \qquad C = \begin{pmatrix} 1 & 0 & 0 \end{pmatrix}.$$

The third coordinate never influences the observed output. The first coordinate alone is not closed, but

$$O_2 = \begin{pmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \end{pmatrix}$$

has rank two and O_3 has the same rank. Proposition 14 gives an exact two-dimensional delay state. Choose $x_0 = (1, 0, q)^\top$ for any q. The first four observations are

$$y_0 = 1, \quad y_1 = 1/2, \quad y_2 = 1/2, \quad y_3 = 11/24.$$

A learner receives those four outputs and the declared delay length $h = 2$. It does **not** receive L, C, O_2 or q. From consecutive observed windows it forms

$$Z_- = \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}, \qquad Z_+ = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 11/24 \end{pmatrix}.$$

Since $\det Z_- = 1/4$, these windows span the declared two-dimensional state. Exact arithmetic gives

$$\boxed{D_2 = Z_+ Z_-^{-1} = \begin{pmatrix} 0 & 1 \\ 1/12 & 5/6 \end{pmatrix}.}$$

The learned recurrence is

$$\boxed{y_{t+1} = \frac{5}{6} y_t + \frac{1}{12} y_{t-1}, \qquad t \geq 1.}$$

It predicts $y_4 = 61/144$, which was not used to identify the coefficients. Proposition 14 and the verified rank condition on the true observation maps establish why this recurrence continues for every initial state of this particular system. The four samples alone establish the fitted recurrence only within the declared closed two-state model class. Proposition 17 forbids confusing those two conclusions.

The reference script separates the roles. One routine fits only the observed samples. A different check, with access to the generating system, verifies the theoretical assumptions and compares subsequent outputs. This prevents hidden access to the true operator from being presented as learning.

The same example connects back to the exact memory kernel. Section 13 expressed the output through a decaying convolution of its history. Here the history is summarized by a two-component recursive state. Neither representation stores the independent third coordinate. Memory and state realization describe the same observable behavior in two different ways.

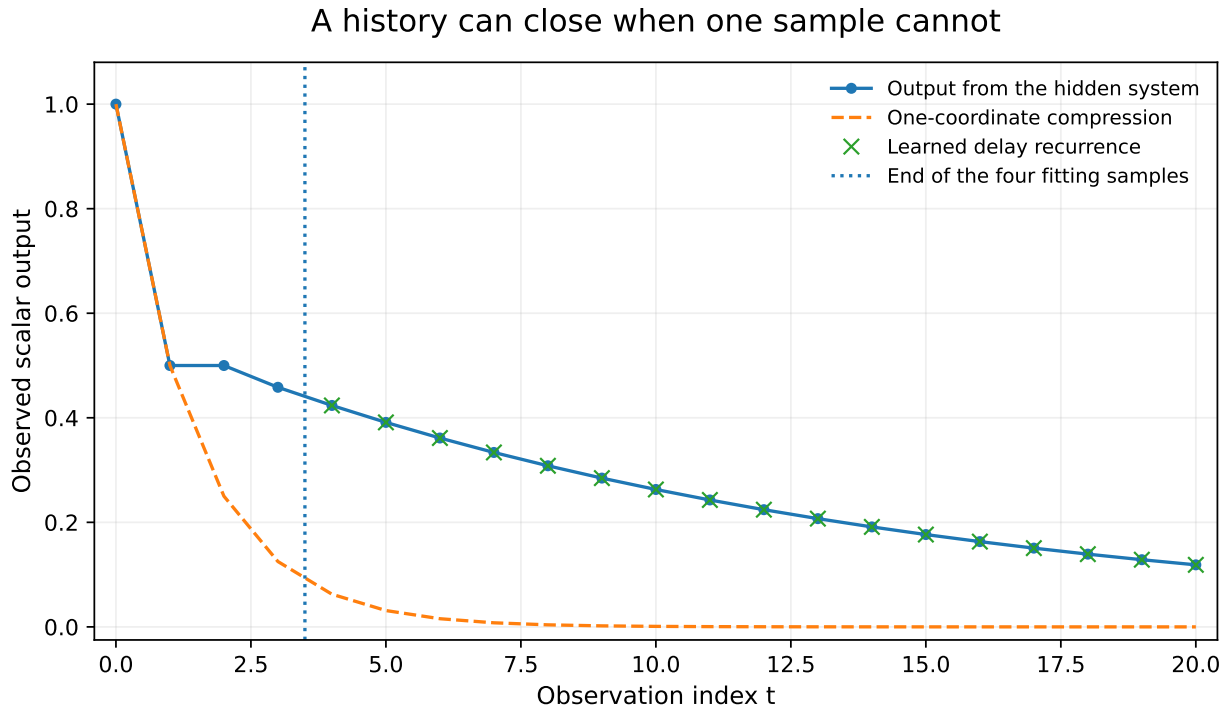


Figure 1: Output-only identification in the declared rational example. Four observations determine the delay recurrence. Later outputs are calculated independently from the hidden system. The one-coordinate compression omits the return path. This is a mathematical example, not a task-suite score.

19. A single propagation law for typed defects

A reconstruction error, an operator-fit error and a refinement error have different origins. Once each has been translated into an error in the state used by the next operation, their propagation can be analyzed by the same identity.

Allow the representation to change with time. Let $\gamma_t : X \to Z_t$ be the declared representation at step t, with a norm on each Z_t. Let $x_{t+1} = F_t(x_t)$ be the true transition and let $\widehat{f}_t : Z_t \to Z_{t+1}$ be the implemented step. Its defect is

$$r_t(x) = \gamma_{t+1}(F_t(x)) - \widehat{f}_t(\gamma_t(x)).$$

This definition already includes a representation switch. It is not restricted to one fixed feature vector.

Proposition 18. Compositional defect propagation. Suppose $\widehat{f}_t$ is λ_t-Lipschitz on a domain containing both $\gamma_t(x_t)$ and the predicted state $\widehat{z}_t$. Suppose $\|r_t(x_t)\| \le \delta_t$ and $\widehat{z}_{t+1} = \widehat{f}_t(\widehat{z}_t)$. Then, with $e_t = \|\gamma_t(x_t) - \widehat{z}_t\|$,

$$\boxed{e_T \le \left(\prod_{j=0}^{T-1} \lambda_j\right) e_0 + \sum_{k=0}^{T-1} \left(\prod_{j=k+1}^{T-1} \lambda_j\right) \delta_k.}$$

Empty products are one. If the output $h_t(x)$ has a decoder H_t with Lipschitz constant κ_t and $\|h_t(x_t) - H_t(\gamma_t(x_t))\| \le \eta_t$, then

$$\|h_t(x_t) - H_t(\widehat{z}_t)\| \le \eta_t + \kappa_t e_t.$$

Proof. Add and subtract $\widehat{f}_t(\gamma_t(x_t))$ to obtain $e_{t+1} \le \delta_t + \lambda_t e_t$. Repeated substitution yields the product-sum formula. Add and subtract $H_t(\gamma_t(x_t))$ for the output inequality. □

For a fixed linear representation R and an estimated transition $\widehat{B}_u$, the defect is $(RL_u - \widehat{B}_u R)x$. For a reference reduced operator B_u, it splits exactly as

$$RL_u - \widehat{B}_u R = (RL_u - B_u R) + (B_u - \widehat{B}_u)R.$$

The first term is representational nonclosure. The second is operator-estimation error. A fit statistic for the second cannot certify that the first vanishes. Proposition 16 supplies a conditional bound on the second in its declared fixed-space setting.

If $\lambda_t \le \rho < 1$ and $\delta_t \le \delta$, then

$$e_T \le \rho^T e_0 + \delta \frac{1 - \rho^T}{1 - \rho}.$$

This is a genuinely recursive prediction bound. It is not merely a one-step comparison made using the true history at every step. If the Lipschitz constants exceed one, the same theorem remains valid but can permit error growth.

19.1 Stability under changing operations

For linear candidate operators, one sufficient way to obtain a common contraction norm is to exhibit a positive-definite matrix M and $0 \le \rho < 1$ with

$$\widehat{B}_u^\top M \widehat{B}_u \preceq \rho^2 M \quad \text{for every declared action } u.$$

Then $\|z\|_M = (z^\top M z)^{1/2}$ gives $\|\widehat{B}_u z\|_M \le \rho \|z\|_M$ for all actions, directly from the displayed inequality. Proposition 18 therefore applies under arbitrary switching among them. This is the common quadratic Lyapunov construction from control theory [R8], used here as an explicit certificate rather than inferred from a list of eigenvalues.

Individual spectral stability does not suffice. The matrices

$$A = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 0 \\ 2 & 0 \end{pmatrix}$$

are both nilpotent, but $BA = \mathrm{diag}(0, 4)$. Alternating them amplifies an appropriate state by four every two steps. Even a single stable nonnormal matrix can produce large transient amplification in a specified norm.

If a learned policy chooses actions from predicted states while the reference process uses actions chosen from true states, the two action sequences may differ. Proposition 18 then needs a bound for that action mismatch, or a Lipschitz bound for the full closed-loop map. The result is not an automatic control-policy guarantee.

20. Truncating memory without losing the prediction bound

Section 8 bounded the omitted tail of the exact linear memory expression using the same resolved history. We now close the separate recursion argument.

Write the exact relation as

$$z_{t+1} = Az_t + \sum_{k=0}^{t-1} K_k z_{t-1-k} + f_t, \qquad K_k = BD^k C, \quad f_t = BD^t q_0.$$

For negative indices set $z_s = 0$, solely as a convention for missing prehistory in this initialized formula. A length-H approximate model, with $H \geq 0$, is

$$\widehat{z}_{t+1} = \widehat{A}\widehat{z}_t + \sum_{k=0}^{H-1} \widehat{K}_k \widehat{z}_{t-1-k}.$$

An empty sum is zero. Let d_t be the exact right-hand side minus this truncated right-hand side evaluated on the **true** history.

Proposition 19. Uniform prediction bound for a stable finite-memory truncation. Suppose

$$a = \|\widehat{A}\| + \sum_{k=0}^{H-1} \|\widehat{K}_k\| < 1, \qquad \sup_{t\geq 0} \|d_t\| \leq \Delta.$$

Let E_0 bound the initialization errors at indices $-H, \ldots, 0$. Then the recursively generated predictions satisfy

$$\boxed{\sup_{t\geq 0} \|z_t - \widehat{z}_t\| \leq \max\left(E_0, \frac{\Delta}{1-a}\right).}$$

If $\|z_t\| \leq Z$ for $t \geq 0$ and $\|D\| \leq \rho < 1$, a valid choice of defect bound is

$$\Delta = Z\left(\|A - \widehat{A}\| + \sum_{k=0}^{H-1} \|K_k - \widehat{K}_k\|\right) + \|B\|\,\|C\| Z \frac{\rho^H}{1-\rho} + \|B\|\,\|q_0\|.$$

The last term can be removed if the exact initialized forcing f_t is retained in the approximate recurrence.

Proof. Subtract the two recurrences and bound every retained history error by the largest of the errors at indices $t-H,\dots,t$. This gives $e_{t+1} \le a \max_{t-H\le s\le t} e_s + \Delta$. The ball of radius $E = \max(E_0, \Delta/(1-a))$ is invariant because $aE + \Delta \le E$. Induction proves the bound. The estimate for Δ separates coefficient errors, the geometric kernel tail and the initialized forcing, using Proposition 10. □

This norm-sum condition is sufficient, not necessary. A stable recurrence with cancellation can fail this test. Failure to find this certificate is not proof of instability.

For the scalar example with $q_0 = 0$, take exact retained coefficients and $z_0 = 1$. All outputs lie in $[0, 1]$: the full nonnegative memory coefficients sum to at most $7/8$, so induction bounds every new output by the previous maximum. Thus $Z = 1$ is valid. With

$$A = 1/2, \quad K_k = \frac{1}{4}\left(\frac{1}{3}\right)^k,$$

one obtains

$$a_H = \frac{7}{8} - \frac{3}{8}3^{-H}, \qquad \Delta_H = \frac{3}{8}3^{-H}, \qquad \boxed{\sup_t |z_t - \widehat{z}_t| \le \frac{3\,3^{-H}}{1+3\,3^{-H}}.}$$

The right-hand side is a theorem for every time in this initialized example. The figure compares it with a separately computed finite-horizon error. The computed maximum is not substituted for the uniform bound.

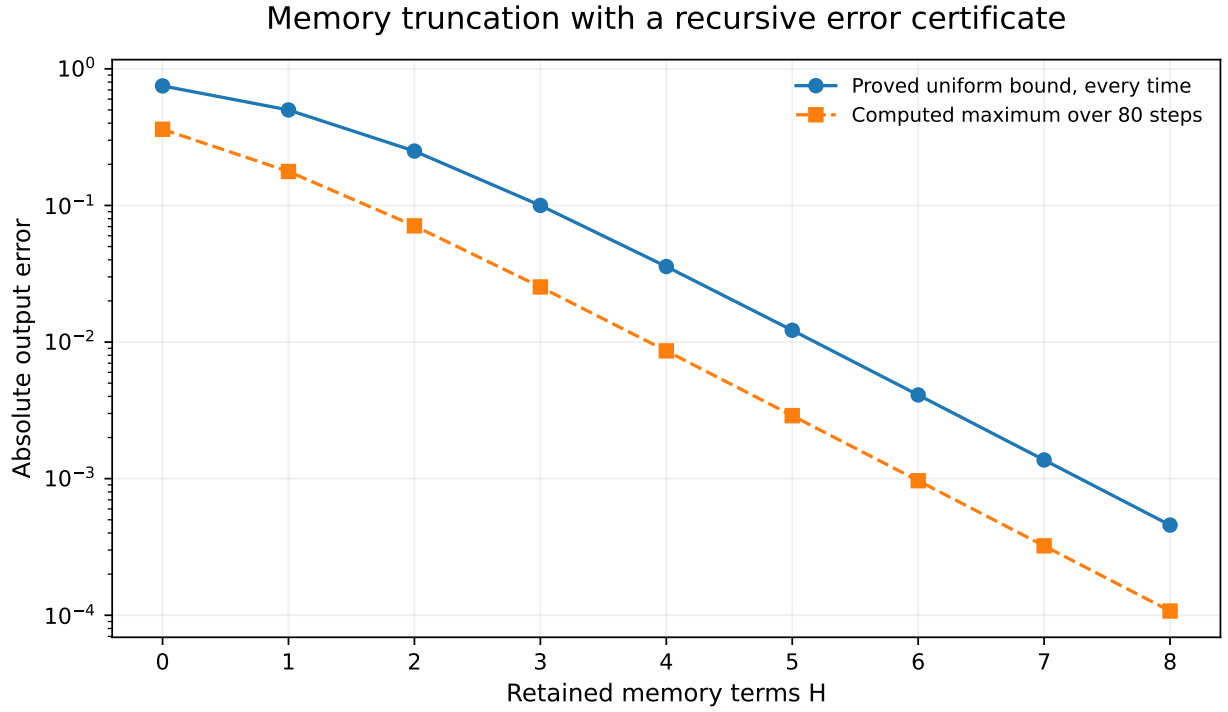


Figure 2: A memory-tail estimate becomes a recursive prediction certificate only after a stability margin is supplied. The measured curve is the maximum over 80 calculated steps. The upper curve is the proved bound for all steps under the stated scalar initialization.

21. Nonlinear packet truncation with an explicit remainder

Polynomial interactions were expanded exactly in Proposition 5. A computational model may retain only some ordered interaction tuples. It then needs a bound on what was omitted, rather than an assumption that packetwise evaluation is exact.

Let P_i be an orthogonal resolution of a finite-dimensional real Hilbert space. Suppose

$$F(x) = b + \sum_{m=1}^{p} B_m(x, \dots, x),$$

with bounded multilinear maps. For each ordered tuple $\boldsymbol{i} = (i_1, \dots, i_m)$, let $\beta_{m,\boldsymbol{i}}$ bound the norm of B_m restricted to $\operatorname{im} P_{i_1} \times \cdots \times \operatorname{im} P_{i_m}$. Let $\mathcal{I}_m$ be the retained tuple set, and let $F_{\mathcal{I}}$ retain the constant term and exactly those interactions.

Proposition 20. Packet-interaction truncation bound. Put $a_i = \|P_i x\|$. Then

$$\boxed{\|F(x) - F_{\mathcal{I}}(x)\| \le \sum_{m=1}^{p} \sum_{\boldsymbol{i} \notin \mathcal{I}_m} \beta_{m,\boldsymbol{i}} \prod_{j=1}^{m} a_{i_j}.}$$

On the ball $\|x\| \le R$, a valid Lipschitz bound for $F_{\mathcal{I}}$ is

$$\operatorname{Lip}(F_{\mathcal{I}}) \le \sum_{m=1}^{p} \sum_{\boldsymbol{i} \in \mathcal{I}_m} m \beta_{m,\boldsymbol{i}} R^{m-1}.$$

Proof. Subtract the retained terms from the exact multilinear expansion. Bound each omitted term by its restricted multilinear norm and sum. For the Lipschitz estimate, telescope the difference between two m-fold evaluations by replacing one argument at a time. Each changed argument has norm at most $\|x - y\|$ after projection, and each unchanged argument has norm at most R. There are m terms for each retained tuple. □

If the declared polynomial itself approximates a nonpolynomial map with remainder at most ε_R on the ball, add ε_R to the defect bound. Without such a remainder estimate, the result is only a theorem for the polynomial, not for the unexpanded physical law.

For a retained state $z = Px$ and omitted state $q = (I - P)x$, dropping every term that uses q gives the coarser majorant

$$\|F(z + q) - F(z)\| \le \sum_{m=1}^{p} \|B_m\| \left((\|z\| + \|q\|)^m - \|z\|^m \right).$$

The tuple-level result is finer because it can distinguish a harmless omitted packet from a strongly coupled one. Ordered tuples must be counted with their multiplicities unless symmetry has explicitly combined them.

A prediction guarantee follows by inserting these defect and Lipschitz bounds into Proposition 18, provided the true and predicted trajectories remain in the declared domain. That domain condition cannot be inferred from a small initial error. One possible construction is to assume that the true map preserves a closed convex set K and to project every approximate output back onto K. Euclidean

projection is nonexpansive and fixes the true output, so it does not increase the one-step approximation bound. This is a declared modification of the predictor, not an identity for an unmodified recurrence.

For the scalar quadratic $(u+v)^2$, retaining only u^2+v^2 omits $2uv$. The packet bound is $2|u||v|$ and is exact in magnitude. Small energy in one packet matters in proportion to both its amplitude and the cross-interaction coefficient.

22. Learning several charts without inventing several worlds

The exact atlas identities constrain the meaning of a fitted model, but independently trained predictors need not satisfy them. Here is a simple coherent reconciliation with a precise objective.

Fix a finite-dimensional represented space X. It may be the observed signal space or a predictive quotient, but it must be the *same* space for all atlases in this section. Let $A_a : X \to C_a$ and $S_a : C_a \to X$ satisfy $S_a A_a = I$. Let $G_a : C_a \to C_a$ be linear fitted maps. Pull them back to object space:

$$L_a = S_a G_a A_a.$$

For positive weights summing to one, define

$$\overline{L} = \sum_a w_a L_a, \qquad \widetilde{G}_a = A_a \overline{L} S_a \quad \text{on } \mathcal{C}_a = \operatorname{im} A_a.$$

Proposition 21. Coherent reconciliation and coordinate-stable transport. The family $\widetilde{G}_a$ is consistent and satisfies every atlas intertwining identity. The operator $\overline{L}$ is the unique minimizer of

$$\sum_a w_a \|L - L_a\|_F^2$$

over linear maps on X, where $\|\cdot\|_F$ is the Frobenius norm. For any target operator L_* and any chosen operator norm,

$$\|\overline{L} - L_*\| \le \sum_a w_a \|L_a - L_*\|.$$

Further, the norm $\|c\|_a = \|S_a c\|$ on $\mathcal{C}_a$ makes every exact transport $T_{ba} = A_b S_a$ an isometry. A represented transition $A_b L S_a$ has exactly the same induced Lipschitz constant in these norms as L has in object space.

Proof. Coherence follows from Proposition 2 because every chart realizes the same $\overline{L}$. Expanding the squares gives

$$\sum_a w_a \|L - L_a\|_F^2 = \|L - \overline{L}\|_F^2 + \sum_a w_a \|L_a - \overline{L}\|_F^2.$$

The weighted cross term vanishes, proving the unique minimizer. The target bound follows from the triangle inequality. The pullback expression is a norm because S_a is inverse to A_a on $\mathcal{C}_a$. Finally, $\|T_{ba} c\|_b = \|S_b A_b S_a c\| = \|S_a c\|$. The same identity for a transition, and the bijection between X and $\mathcal{C}_a$, establish equality of Lipschitz constants. □

This reconciliation is ordinary weighted least squares in a common semantic space. It does not prove that consensus is correct, reduce statistical error under arbitrary dependence, or turn deterministic recodings into independent evidence. The output must also be reanalyzed into a realizable coefficient vector. An independently fitted redundant output can have an additional inconsistency residual $(I - A_a S_a)G_a A_a x$ before reconciliation.

For Euclidean coefficient errors, conditioning remains important. If A_a has smallest singular value at least $m_a > 0$ and largest singular value at most M_a, its canonical synthesis $A_a^\dagger$ has norm at most $1/m_a$, and $\|T_{ba}\| \le M_b/m_a$. An arbitrary left inverse can behave much worse away from the consistent coefficient space, so the canonical-synthesis qualification matters.

A learned atlas can therefore be parameterized within prescribed singular-value bounds, with synthesis tied to its inverse or pseudoinverse. This gives a construction contract. It is not a claim that every unconstrained training procedure respects those bounds.

When atlases change over time but represent the same state, using the pullback norms avoids multiplying arbitrary chart condition numbers at every step. Exact transport telescopes to the direct transport. Numerical synthesis errors and genuinely inconsistent predictions remain defects and must still be bounded.

23. Refinement cannot manufacture missing information

An atlas change is an invertible recoding of the same represented information. Predictive refinement usually adds information. Treating these as identical operations would undo the central distinction of the foundation.

Proposition 22. The upward-lift obstruction. Let $\gamma : X \to Z$ be a coarse representation and $\eta : X \to W$ a proposed refined representation. There is a deterministic map $J : \gamma(X) \to \eta(X)$ satisfying $\eta = J\gamma$ if and only if

$$\gamma(x) = \gamma(x') \implies \eta(x) = \eta(x') \quad \text{for every } x, x'.$$

For linear representations, this is equivalent to $\ker \gamma \subseteq \ker \eta$. If two states share a coarse representation but their refined representations differ, any predicted refined value w has worst-case error at least

$$\max(\|w - \eta(x)\|, \|w - \eta(x')\|) \ge \tfrac{1}{2}\|\eta(x) - \eta(x')\|.$$

Proof. A lift must be constant on each coarse fiber. Conversely, the stated condition defines it unambiguously on every fiber. The linear version is the kernel factorization criterion. The lower bound follows from the triangle inequality between the two refined values through w. □

For a strict refinement, the coarse state is recoverable from the refined one but not conversely. In the linear case this typically means $\ker \eta \subsetneq \ker \gamma$, exactly the opposite of the condition required for an upward lift.

For example, $\gamma(x_1, x_2) = x_1$ and $\eta(x_1, x_2) = (x_1, x_2)$. The states $(0, 1)$ and $(0, -1)$ have the same coarse state, but any common point initialization has worst-case error at least one in Euclidean norm. Padding a vector with zeros is a choice, not information recovery.

A refinement step therefore needs one of three explicitly distinguished inputs: an archive that already contains the relevant observations, new observations, or a set-valued or probabilistic state retaining

uncertainty about the missing details. An exact archive of measured outputs can initialize a longer history when those samples were stored. It cannot recover a latent coordinate that never affected any measurement.

For adaptive prediction, an attempted lift J_t contributes its own initialization defect

$$\eta_t(x_t) - J_t\gamma_t(x_t).$$

It belongs in Proposition 18's initial error or in the defect of the composed transition. It cannot be set to zero merely because a finer packet tree has been allocated.

A data-dependent representation schedule also requires care. The pathwise bound applies to the realized schedule when all of its hypotheses hold along that schedule. Comparing two different schedules chosen by true and predicted histories introduces a selection discrepancy. No uniform statistical consistency of adaptive selection follows solely from the deterministic propagation law.

24. A growing operator grammar with coherent evidence

A large algebraic signal model should be able to add operator programs without changing the meaning of its existing evidence. This requires more than assigning a new softmax weight when a candidate is introduced.

Let $\mathcal{G}$ be a countable declared program set, with one fixed probability mass function $\pi(g) > 0$ satisfying $\sum_g \pi(g) = 1$. Every program specifies a likelihood $L_g(D)$ for the **same** observed data event or density relative to a common base measure. Assume

$$0 < Z(D) = \sum_{g\in\mathcal{G}} \pi(g)L_g(D) < \infty.$$

Let finite active sets $\mathcal{G}_n$ increase to $\mathcal{G}$. Their unnormalized evidence is $Z_n = \sum_{g\in\mathcal{G}_n} \pi(g)L_g(D)$. Only stages with $Z_n > 0$ are considered.

Proposition 23. Exact accounting for an enumerated grammar. Define the full posterior $P(g \mid D) = \pi(g)L_g(D)/Z$ and extend the active posterior $P_n(g \mid D) = \pi(g)L_g(D)/Z_n$ by zero outside $\mathcal{G}_n$. Then

$$\boxed{\|P - P_n\|_{\mathrm{TV}} = \tau_n, \qquad \tau_n = 1 - \frac{Z_n}{Z}.}$$

For any common family of candidate predictive probability kernels, the corresponding predictive mixtures differ in total variation by at most τ_n. If the omitted unnormalized evidence has a bound

$$\sum_{g\notin\mathcal{G}_n} \pi(g)L_g(D) \le b_n,$$

then

$$\tau_n \le \frac{b_n}{Z_n + b_n}.$$

Proof. On the active set, $P = (Z_n/Z)P_n$, so the total mass of the difference there is τ_n. The mass outside is also τ_n. Half the sum of absolute differences is therefore τ_n. Applying the same probability kernel to two mixing measures cannot increase total variation: integrate any event probability, a function taking values in $[0, 1]$, against their difference. For the final bound, write $Z = Z_n + W_n$ with $0 \le W_n \le b_n$ and use that $W_n/(Z_n + W_n)$ increases with W_n. □

Since the active sets exhaust the grammar and the evidence sum is finite, $Z_n \uparrow Z$ and $\tau_n \downarrow 0$. This is posterior approximation *inside the declared grammar*. It does not establish that the real signal-generating mechanism belongs to it.

For a discrete data sequence, a probability likelihood is at most one, so the omitted prior mass is a valid b_n. A continuous-data density need not be at most one. It requires a density envelope or another justified bound on omitted evidence. A small unnormalized tail can also carry substantial posterior mass when the active evidence is very small.

The fixed prior need not be uniform. For an explicit enumeration $g_1, g_2, \ldots$, $\pi(g_k) = 2^{-k}$ is a normalized example, with omitted prior mass 2^{-n} after n programs. It expresses a particular enumeration bias, not a canonical prior on mathematical meaning.

A program outside the enumerated prefix can be accounted for only when its prior and likelihood are defined or bounded. A genuinely outside-grammar component needs its own model. Assigning an unexplained residual probability is not a Bayesian treatment of missing mechanisms.

24.1 Refining a program without double counting it

A representation-rich language can contain many syntactic versions of one mechanism. It can also split one broad mechanism into more specific children. These operations must conserve prior mass when they are intended as a re-expression of the same model.

Proposition 24. Prior-preserving program refinement. Suppose a parent program g is replaced by children g_j with weights $\alpha_j \ge 0$, $\sum_j \alpha_j = 1$, and suppose its likelihood is the mixture

$$L_g(D) = \sum_j \alpha_j L_{g_j}(D).$$

Give child j prior mass $\pi(g)\alpha_j$ and leave all other programs unchanged. Then total evidence is unchanged and the sum of the children's posterior masses equals the former parent's posterior mass. Posterior predictive distributions are also unchanged when the same mixture relation holds for the joint likelihood of (D, y) for every predicted event y.

Proof. The children's total unnormalized evidence is $\sum_j \pi(g)\alpha_j L_{g_j}(D) = \pi(g)L_g(D)$. Thus both the total normalizer and every other program's posterior are unchanged. The same sum for joint evidence gives the predictive statement after normalization by the unchanged evidence of D. □

The mixture identity must hold for the **whole data record**. Averaging one-step likelihoods independently at each time generally describes a switching mechanism, not a fixed child selected once for the whole sequence.

If the children are only exact atlas recodings of the same operator and data model, their likelihoods are identical. Splitting the original prior mass leaves every prediction unchanged. Giving each duplicate a fresh full prior mass instead changes the model's prior preference. That may be an intentional new prior, but it is not invariance under representation.

The implementation's existing held-out predictive weights, described in Section 10, are not replaced by these identities merely by calling them posterior weights. The proposition supplies the probabilistic semantics that a future grammar-weighting implementation would need to enforce.

25. The finite learned-realization theorem

The preceding results combine into a concrete completion of one part of the learning problem. This is the strongest claim made for incomplete observations in this volume.

Proposition 25. Observationally grounded finite linear realization. Assume all of the following:

1. The data come from an autonomous finite-dimensional linear process with declared outputs $y_t = Cx_t$.
2. A delay length h satisfies $\operatorname{rank} O_{h+1} = \operatorname{rank} O_h$.
3. Exact observed input windows in Z_- span $V_h = \operatorname{im} O_h$, and Z_+ records their true one-step successor windows.

Then an algorithm using only (Z_-, Z_+) recovers the unique transition on V_h. Initialized with a fully observed length-h window, it predicts every later output exactly. Any lossless atlas of V_h, any complete packet decomposition of that space, and every coherently transported realization of the learned transition preserve the same predictions. Hidden variables outside the observable quotient need not be recovered.

If the fixed-space assumptions of Proposition 16 replace exact observations, that proposition bounds operator error. Whenever the resulting transition defects and Lipschitz constants satisfy Proposition 18, its product-sum bound controls the output prediction error. A common contraction margin yields the stated uniform bound.

Proof. Proposition 14 gives the unique true transition D_h on V_h. Because the observed windows span V_h, Proposition 15 identifies exactly its restriction, which is the entire required operator. Equality of the learned and true states follows by induction after initialization. The current output is the last block of the history state. Propositions 1–4 preserve state reconstruction, transport and complete decomposition, so they preserve that output sequence. The perturbation statement is the direct composition of Propositions 16 and 18 with the declared output decoder. □

This establishes an actual route from observations to a composable algebraic signal model under explicit finite assumptions. It is more than a fit on a few points. It is also less than unrestricted nonlinear world-model discovery. In particular, the algorithm does not infer assumptions 1–3 merely by returning a matrix. The accompanying certificate keeps the data-verified identities separate from those model assumptions.

26. Second-block checkpoint and remaining questions

The known-operator constructions in Part I now have an observation-based continuation. There is an exact causal realization theorem for a finite linear observable sector, a data-span identifiability criterion and a fixed-space perturbation bound. Recursive memory truncation has a sufficient stability theorem. Polynomial packet omissions have a remainder bound. Learned chart descriptions can be reconciled, refinement has an explicit information requirement, and grammar growth has an evidence-preserving semantics.

The theoretical center is a sequence of obligations:

$$\begin{aligned} &\text{accessible observations} \longrightarrow \text{declared predictive state} \longrightarrow \text{identified operations},\\ &\text{typed defects} \longrightarrow \text{stable composition} \longrightarrow \text{coherent refinement and inference}. \end{aligned}$$

A training residual cannot stand in for coverage of unobserved directions. A small coefficient error cannot stand in for well-conditioned synthesis. A posterior within a grammar cannot stand in for a probability that the grammar is complete.

26.1 The remaining research questions are narrower

Proposition 20 controls omissions in a declared polynomial expansion. It does not discover an invariant nonlinear dictionary or prove that its dimension stays finite. Selection and stability of nonlinear invariant features remain separate tasks [R6].

Stochastic partially observed processes need their own accessible state construction, positive probability semantics and finite-sample control. The HMM literature gives examples under rank and separation assumptions [R7], not guarantees for an arbitrary operator grammar. Noisy feature-space learning needs subspace perturbation and rank-selection analysis. Dependent sampling and active observations need their own statistical and excitation assumptions.

Without contraction or another stability certificate, the product-sum error bound remains valid but need not stay small. Unbounded infinite-dimensional operators still need domain control. Adaptive selection of representations still needs consistency beyond a pathwise deterministic estimate. These questions delimit the present theorems rather than invalidate their stated finite conclusions.

26.2 Release and verification status

The preceding foundations edition 0.2.0 contained Propositions 1–25, including twelve statements added in Part II. Every added statement is proved in the text. The accompanying rational, symbolic and separately labeled numerical calculations check finite examples and counterexamples. No general Lean kernel verification is claimed.

The 2.3.0 runtime is unchanged. The output-only reference learner implements data-span identification, not unrestricted discovery. Source lineage, assumptions, implementation boundaries, calculation reports and the complete preceding foundations archive accompany this edition.

The contribution is a connected foundation and explicit special-case repairs. Observability, predictive states, least squares, invariant subspaces, Lyapunov stability and Bayesian mixture identities have classical antecedents. The proposition count is not a count of historical discoveries.

Part III. Controlled stochastic realization

The next step is not to rename a deterministic predictor as a probabilistic one. A stochastic signal model must say what its state predicts, which actions intervene on the process, how observations update the state, and why every resulting number is a probability. These are separate obligations. A low dimensional linear representation can satisfy its reconstruction equations while failing probability positivity.

Part II established observable learning for an autonomous linear signal. Here the observable objects are probabilities of finite action and observation experiments. We derive a finite realization from them, show when it is minimal, give an explicit probability-valid state domain, and separate exact reconstruction from noisy estimation. The final construction connects back to Komorebi's packets through a normalized closure-defect identity.

The classical antecedents are predictive state representations, observable operator models and finite-rank realization [R5, R7, R9]. Those traditions already use probabilities of tests and products of operators. The purpose here is to make their assumptions and consequences explicit within the realization calculus. The twelve propositions below are a connected development, not twelve claims of historical novelty. No statement here replaces a proof with a numerical check.

27. Actions and observations are different inputs

Let $\mathcal{A}$ and $\mathcal{O}$ be finite nonempty action and observation sets. A symbol $s = (a, o)$ records an action followed by an observation. A word $w = s_1 \cdots s_k$ has temporal order from left to right. The empty word is ϵ.

Write

$$f(w) = \Pr(o_1, \ldots, o_k \mid \mathrm{do}(a_1, \ldots, a_k)).$$

The process is reset to one specified initial distribution before this experiment. The actions in this definition are fixed externally. This is not, in general, the distribution obtained by conditioning a passive record on an adaptively selected sequence of actions. We assume the consistency laws

$$f(\epsilon) = 1, \qquad f(w) \geq 0, \qquad \sum_{o \in \mathcal{O}} f(w(a, o)) = f(w) \quad \text{for every } w, a.$$

These laws specify causal probabilities on finite observation trees. A fixed history-dependent policy can be evaluated by selecting the action on each branch and multiplying the appropriate conditional probabilities. This does not establish optimality of that policy or its identification from insufficient data.

27.1 A finite hidden-state construction

A concrete realization has a belief $p \in \mathbb{R}^n_+$ with $\mathbf{1}^\top p = 1$. For each $s = (a, o)$, let $M_s \in \mathbb{R}^{n \times n}$ be entrywise nonnegative, with

$$\mathbf{1}^\top \sum_o M_{a,o} = \mathbf{1}^\top.$$

The matrix entry $M_{a,o}(j, i)$ is the probability of next hidden state j and observation o, given current hidden state i and chosen action a. This is a discrete probability instrument. It includes an action-specific transition followed by an observation, but does not require that factorization.

Put $M_w = M_{s_k} \cdots M_{s_1}$. Then

$$f(w) = \mathbf{1}^\top M_w p_0, \qquad p^+ = \frac{M_s p}{\mathbf{1}^\top M_s p}$$

on branches of positive probability. An observation with zero probability has no conditional posterior under that model. It must not be handled by silently dividing by an arbitrary numerical floor.

Proposition 26. Finite instrument closure. Starting with $W_0 = \mathrm{span}\{\mathbf{1}^\top\}$, define

$$W_{j+1} = W_j + \sum_{s \in \mathcal{A} \times \mathcal{O}} W_j M_s.$$

This stabilizes after at most $n - 1$ strict enlargements at

$$W_* = \mathrm{span}\{\mathbf{1}^\top M_w : w \text{ a finite word}\}.$$

Choose a full row rank matrix R with row space W_* and first row $\mathbf{1}^\top$. There are unique matrices B_s with

$$RM_s = B_s R, \qquad \lambda = e_1^\top, \qquad \lambda R = \mathbf{1}^\top.$$

For $z = Rp$, both the next-observation likelihood and the posterior predictive state close:

$$\Pr(o \mid z, a) = \lambda B_{a,o} z, \qquad z^+ = \frac{B_{a,o} z}{\lambda B_{a,o} z}.$$

Every linear summary determining all word probabilities by linear readouts for every initial belief has row space containing W_*.

Proof. The nested-space argument from Proposition 8 applies to the finite family M_s. Multiplication of a test row by M_s prepends the symbol s to that test, so the stable span is exactly the displayed word span. Invariance supplies the matrices B_s, and full row rank makes them unique. Apply R to the unnormalized and normalized hidden-state updates. If all test probabilities are linear functions of another summary, then each row $\mathbf{1}^\top M_w$ lies in its row space. Equality on all beliefs implies equality on their linear span, which is $\mathbb{R}^n$. □

This proposition uses a known hidden-state model. It is not yet an observable-data learner. It also concerns all initial beliefs, whereas a model initialized at one fixed p_0 can have a smaller reachable predictive dimension. The next construction handles that distinction without requiring recovery of hidden states.

28. The state is a function of possible future experiments

For a history h, define its unnormalized future-behavior column

$$v_h(t) = f(ht),$$

where t ranges over every finite test word. Form the infinite history/test matrix

$$\mathscr{H}_{t,h} = f(ht).$$

Its rank means the dimension of the span of its columns, equivalently the supremum of the ranks of finite submatrices. The columns are functions, not vectors obtained by physically observing every future. Their finite coordinate realization becomes learnable only when enough experiment probabilities are available.

Proposition 27. Minimal finite behavior realization. If $\operatorname{rank} \mathscr{H} = r < \infty$, the vector space $V = \operatorname{span}\{v_h\}$ has dimension r. There are well-defined linear maps $B_s : V \to V$, an initial vector v_ϵ, and a linear mass functional $\lambda(v) = v(\epsilon)$ satisfying

$$B_s v_h = v_{hs}, \qquad f(s_1 \cdots s_k) = \lambda B_{s_k} \cdots B_{s_1} v_\epsilon.$$

Every linear realization of f has dimension at least r. Any two reachable and observable realizations of dimension r differ by an invertible change of coordinates. For $f(h) > 0$, the normalized state $v_h/f(h)$ determines every future test probability conditional on h and the declared future interventions.

Proof. If $\sum_h c_h v_h = 0$, then for every test t,

$$\sum_h c_h v_{hs}(t) = \sum_h c_h f(hst) = 0,$$

because st is another test. Thus shifting columns preserves all linear relations and defines B_s. The probability formula follows by induction. For any d-dimensional realization, the history/test matrix factors into test rows and reachable state columns, so its rank is at most d.

For uniqueness, map the reachable vector after each history in one minimal realization to the corresponding vector in the other. If a linear combination vanishes in the first, every future test evaluates to zero on it. Those same evaluations vanish in the second, where observability makes the combination zero. The map is therefore well-defined and bijective, and intertwines every symbol operator and both boundary vectors. Finally $f(ht)/f(h)$ is exactly the conditional experiment probability. □

The coordinate dimension r includes probability mass. Normalized states lie in the affine hyperplane $\lambda z = 1$, so their affine dimension is at most $r - 1$. A finite hidden-state model of size n has behavior rank at most n. The theorem itself requires finite behavior rank, not a preferred hidden-state ontology.

Two realized histories with equal normalized future columns are equivalent for every declared future test. This is a predictive equivalence, not equality of their physical pasts or lossless archives.

29. Recovering the operators from an observable probability table

Select r tests $t_1, \ldots, t_r$ and r histories $h_1, \ldots, h_r$, with $t_1 = \epsilon$, for which the square matrix

$$H_{ij} = f(h_j t_i)$$

is invertible. Such a choice exists under the rank hypothesis of Proposition 27. The nonzero empty-test row can be extended to a row basis. Select independent history columns for that basis.

For each symbol s, also obtain the shifted table and initial test vector

$$(G_s)_{ij} = f(h_j s t_i), \qquad (b_0)_i = f(t_i).$$

Proposition 28. Observable table reconstruction. Assume the full behavior rank is at most r and the displayed $r \times r$ matrix H is invertible. Then the full rank is exactly r, and

$$\boxed{B_s = G_s H^{-1}, \qquad \lambda = e_1^\top, \qquad b_0 = (f(t_i))_i}$$

is a minimal realization of every word probability. Its state after history h is

$$b(h) = (f(h t_i))_i = B_h b_0.$$

The normalized state is $b(h)/f(h)$ when $f(h) > 0$. Its update uses the formula in Proposition 26, with no hidden-state coordinates supplied to the reconstruction algorithm.

Proof. Invertibility gives rank at least r, so the assumed upper bound is attained. Evaluation on the selected tests is an isomorphism $V \to \mathbb{R}^r$, because it maps the chosen basis of history columns to the invertible matrix H. The shifted images of those basis columns are precisely the columns of G_s, so $B_s H = G_s$. The first selected test evaluates the mass. Initial evaluation gives b_0. Proposition 27 supplies correctness for every history, not only the selected columns. □

There are two ways to justify coverage in this statement. One can know that the selected row and column families span the true behavior space. Alternatively, a justified upper bound r on the process rank, together with the invertible $r \times r$ block, proves that coverage. The observed invertibility by itself only gives a lower bound. An unknown higher-rank process can agree on a finite table and differ later.

Exact experiment probabilities are population quantities. The reference example below supplies rational tables from a declared process and keeps that table generator separate from the learner. Section 36 supplies one finite sampling protocol for estimating those entries. Neither a hidden matrix nor its probability cone is secretly passed to the observable learner.

30. Probability validity is not a rank condition

A real matrix product can yield a negative number. Therefore a fitted factorization is not a stochastic model merely because it is low rank or exactly interpolates a probability table. Spectral estimators can encounter negative-probability problems in practice [R10]. We address the structural issue directly.

In the canonical behavior space V, define

$$\mathcal{C} = \{v \in V : v(t) \geq 0 \text{ for every finite test } t\}.$$

Proposition 29. Canonical probability cone. The set $\mathcal{C}$ is a closed convex pointed cone that spans V. Every B_s preserves it. The mass functional is strictly positive on its nonzero elements, and

$$\sum_o \lambda B_{a,o} = \lambda.$$

Consequently the normalized base

$$\mathcal{K} = \{v \in \mathcal{C} : \lambda v = 1\}$$

is a compact convex probability-valid state space. Every positive-probability branch maps $\mathcal{K}$ into itself by normalized updating. In the core-test coordinates of Proposition 28, $\mathcal{K} \subseteq \{1\} \times [0,1]^{r-1}$.

Proof. Each test evaluation is a continuous linear functional on finite-dimensional V, so the cone is closed and convex. If both v and $-v$ belong to it, every test value is zero and $v = 0$. It spans V because it contains every column v_h. Also $(B_s v)(t) = v(st)$, proving invariance.

The consistency law $\sum_o v(t(a,o)) = v(t)$ holds for each history column and hence throughout V. Iterating it along any fixed action schedule expresses $v(\epsilon)$ as a sum of nonnegative values for all observation words on that schedule. If $v(\epsilon) = 0$, each summand is zero, for every schedule and length, so $v = 0$. The same sums show $0 \leq v(t) \leq 1$ when $\lambda v = 1$. They also give mass conservation. The base is closed and bounded in the finite core coordinates and thus compact. Invariance and positive normalization prove the posterior assertion. □

This cone is defined by all finite tests. Its existence is not an algorithm for finding a finite list of inequalities. A finite list gives an outer approximation and can miss a longer-test violation. The cone also need not equal the convex hull of the states reachable from the fixed initial distribution.

A practical sufficient certificate can instead use a known closed cone C_0 with nonempty normalized base, a strictly positive mass functional, $b_0 \in C_0$, $B_s C_0 \subseteq C_0$, and mass conservation. A nonnegative hidden-state realization supplies one after a coordinate change. Such a certificate may use information unavailable to an observable-only learner and must be labeled accordingly.

Negative entries of B_s do not, by themselves, prove invalidity. Coordinates can be signed while the represented state remains in the correct cone. Conversely, clipping negative coefficients can destroy the operator identities. Validity belongs to the pair consisting of the operators and their admissible cone, not to a coordinatewise heuristic.

31. Packets can reconstruct a belief without being beliefs

Let $P_1, \dots, P_m$ be a complete family of linear packet maps on a predictive space, with $\sum_i P_i = I$. Algebraic reconstruction gives $z = \sum_i P_i z$. It does not imply that $P_i z$ is an unnormalized probability state.

Proposition 30. Positive packet decomposition and its boundary. Suppose a cone C has a strictly positive mass functional λ. If every packet map preserves C, then each normalized $z \in C$ decomposes as a genuine mixture

$$z = \sum_{i:w_i>0} w_i z_i, \qquad w_i = \lambda P_i z \geq 0, \qquad z_i = P_i z / w_i, \qquad \sum_i w_i = 1.$$

A zero weight has $P_i z = 0$. In the standard cone $\mathbb{R}^n_+$, however, a complete family of cone-preserving idempotents must consist of diagonal coordinate-selection matrices. In particular, a nontrivial Fourier or Haar packet family cannot generally be reinterpreted as a mixture of probability states merely because it is orthogonal and complete.

Proof. Cone preservation and strict positivity give the weights and the zero-weight claim. Apply λ to the complete decomposition to obtain their sum. Normalization of each nonzero term gives the mixture.

On $\mathbb{R}^n_+$, cone preservation is entrywise nonnegativity. Since the matrices sum to I, every off-diagonal entry in every matrix is zero. Idempotence forces each diagonal entry to be zero or one, and completeness makes the selections disjoint and exhaustive. □

For a two-state belief $p = (3/4, 1/4)^\top$, the Haar detail projector

$$P_{\mathrm{d}} = \frac{1}{2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

produces $(1/4, -1/4)^\top$. It is a valid signed detail packet, but not a subprobability distribution. Its total mass is zero despite being nonzero, which cannot happen inside a cone with a strictly positive mass functional.

This does not prohibit useful spectral packets. It fixes their type. Signed algebraic packets can be used for transfer, approximation and diagnostics. They must be synthesized before probability interpretation unless a separate positive decomposition has been established. That distinction prevents a reconstruction identity from being misused as a Bayesian mixture identity.

32. Action coverage is a mathematical requirement

Proposition 31. Missing-intervention nonidentifiability. There are finite controlled stochastic models that agree on every possible observation record generated using only action a, but disagree under another action b. Therefore even infinitely many passive records confined to a cannot identify the unrestricted controlled model.

Proof. Use one hidden state and two observations. In both models, action a gives probabilities $(1/2, 1/2)$. Under b, let the first model give $(1, 0)$ and the second $(0, 1)$. Every length-k record under a has probability 2^{-k} in both models. Their first observation under b differs with probability one. □

This is a coverage obstruction, not an optimization problem. It remains even when the hidden dimension is known exactly. Richer transforms of the same action-a observations do not create data about b.

Action selection based on past observations requires a further distinction. Under a known policy π, an action/observation word has joint probability

$$\Pr_{\pi}(w) = f(w) \prod_{j=1}^{|w|} \pi(a_j \mid h_{j-1}),$$

provided the action randomization has no access to hidden information beyond the declared history. The policy factors depend on the observed branch. Conditioning on a selected action sequence generally reweights branches rather than removing these factors. Recovering intervention probabilities from such data requires an appropriate coverage and reweighting argument. The reset experiments in Section 36 avoid that additional inference problem.

33. Learning the coordinate space under bounded noise

Part II bounded operator error after a state space and basis were fixed. Here we also bound one step of data-dependent space selection. This remains a finite-matrix statement. It does not certify that a chosen finite history/test table captures the full process.

Let $X \in \mathbb{R}^{m \times n}$ have rank $r \geq 1$, let $s = \sigma_r(X) > 0$, and observe $Y = X + E$ with $\|E\|_2 \leq \varepsilon < s/2$. Let P project onto $\operatorname{im} X$, and let $\widehat{P}$ project onto the leading r left singular vectors of Y.

Proposition 32. Finite-table rank separation and subspace stability. One has

$$\sigma_r(Y) \geq s - \varepsilon, \qquad \sigma_{r+1}(Y) \leq \varepsilon, \qquad \|P - \widehat{P}\|_2 \leq \frac{2\varepsilon}{s} < 1.$$

The $(r+1)$st singular value is interpreted as zero when r is the maximum possible rank. If U and $\widehat{U}$ are orthonormal bases of the two ranges, then

$$\sigma_{\min}(\widehat{U}^{\top} U) \geq \sqrt{1 - (2\varepsilon/s)^2}.$$

Thus $x \mapsto \widehat{U}^{\top} x$ is injective on the true column space. Its inverse on that space is

$$y \longmapsto U(\widehat{U}^{\top} U)^{-1} y,$$

with norm at most $[1 - (2\varepsilon/s)^2]^{-1/2}$.

Proof. The min-max characterization of singular values and the triangle inequality imply that adding E changes each singular value by at most ε. The leading-r singular subspace gives $\|(I - \widehat{P})Y\|_2 = \sigma_{r+1}(Y) \le \varepsilon$. Hence

$$\|(I - \widehat{P})X\|_2 \le 2\varepsilon, \qquad \|(I - \widehat{P})P\|_2 = \|(I - \widehat{P})XX^\dagger\|_2 \le 2\varepsilon/s.$$

For equal-rank orthogonal projections, the latter norm equals $\|P - \widehat{P}\|_2$. To see the equality, diagonalize $U^\top \widehat{U}$ by singular value decomposition. Each nontrivial paired principal plane contributes a sine to both norms, while common directions contribute zero. On the same planes the singular values of $\widehat{U}^\top U$ are the corresponding cosines. This proves the last bound and the inverse formula. □

A threshold τ with $\varepsilon < \tau < s - \varepsilon$ recovers the rank of this finite matrix. The unknown true gap must be justified or replaced by a valid confidence interval. Rank thresholds do not exclude modes absent from the selected tests or unexcited by the selected histories.

The inverse formula contains the unknown true basis U. It proves identifiability and conditioning relative to that space, not that a learner already knows a perfect decoder. When using the estimated subspace as coordinates for a realization, all boundary vectors, symbol operators and probability constraints must be transformed together. Signed singular-vector coordinates are not new independent observations.

34. Noisy operator reconstruction in a fixed core

After choosing a core block, the square reconstruction formula yields a particularly direct error identity. It is useful even when the core was chosen on a separate discovery data set.

Let $G_s = B_s H$ with $s_H = \sigma_{\min}(H) > 0$. Observe

$$\widetilde{H} = H + E, \qquad \widetilde{G}_s = G_s + F_s, \qquad \|E\|_2 < s_H.$$

Proposition 33. Noisy core-block realization. The estimate $\widehat{B}_s = \widetilde{G}_s \widetilde{H}^{-1}$ satisfies

$$\widehat{B}_s - B_s = (F_s - B_s E)\widetilde{H}^{-1},$$

and therefore

$$\boxed{\|\widehat{B}_s - B_s\|_2 \le \frac{\|F_s\|_2 + \|B_s\|_2 \|E\|_2}{s_H - \|E\|_2}.}$$

A bound using only the fitted operator and the same perturbation budgets is

$$\|\widehat{B}_s - B_s\|_2 \le \frac{\|F_s\|_2 + \|\widehat{B}_s\|_2 \|E\|_2}{s_H}.$$

If $\widetilde{s} = \sigma_{\min}(\widetilde{H})$ is known and $\|E\|_2 \le e < \widetilde{s}$, the last denominator may be replaced by $\widetilde{s} - e$.

Proof. Since $\widetilde{H}$ is invertible, subtract B_s from its fitted expression and collect the terms. The least singular value of $\widetilde{H}$ is at least $s_H - \|E\|_2$. For the second identity, use

$$(\widehat{B}_s - B_s)H = F_s - \widehat{B}_s E$$

and multiply by H^{-1}. Finally $s_H \ge \widetilde{s} - e$ by singular-value perturbation. □

When the first test is empty, the mass row $\lambda = e_1^\top$ is fixed rather than estimated. Set the first component of the initial vector to one as required by $f(\epsilon) = 1$. Errors in other initial test probabilities remain initial-state error. This convention removes an avoidable normalizer-estimation ambiguity but does not enforce cone preservation or mass conservation for the fitted symbol family.

Numerical implementation should solve the linear systems rather than explicitly form a floating-point inverse. The theorem describes the exact algebraic estimator. Finite-precision solve error is an additional residual, which the reference script measures separately.

35. Conditioning a prediction can amplify its error

Let C be a probability cone with compact normalized base K and mass row λ. Work in a declared norm and its dual. Suppose true normalized states have norm at most R. Let a true branch have

$$u = B_s z, \qquad p = \lambda u \ge \eta > 0, \qquad z^+ = u/p \in K.$$

For a predicted state $\widehat{z}$, form $\widehat{u} = \widehat{B}_s\widehat{z}$, $\widehat{p} = \lambda\widehat{u}$, and define $d = \widehat{u} - u$. Write $e = \|\widehat{z} - z\|$, $\|\widehat{B}_s\| \le L$, and $\|\widehat{B}_s - B_s\| \le \epsilon_B$. Then

$$\|d\| \le Le + \epsilon_B R =: D.$$

Proposition 34. Normalization-aware filtering stability. If $\|\lambda\|_* D < \eta$, then $\widehat{p} > 0$ and

$$\widehat{z}^+ - z^+ = \frac{(I - z^+\lambda)d}{\widehat{p}},$$

so

$$\boxed{\|\widehat{z}^+ - z^+\| \le \frac{(1 + R\|\lambda\|_*)D}{\eta - \|\lambda\|_* D}.}$$

Alternatively, if each realized fitted denominator is checked to satisfy $\widehat{p}_t \ge \widehat{\eta}_t > 0$, the pathwise recursion obeys

$$e_{t+1} \le \alpha_t e_t + \beta_t, \qquad \alpha_t = \frac{(1 + R\|\lambda\|_*)L_t}{\widehat{\eta}_t}, \qquad \beta_t = \frac{(1 + R\|\lambda\|_*)\epsilon_{B,t}R}{\widehat{\eta}_t}.$$

The product-sum bound of Proposition 18 applies. A uniform bound follows if these constants have a common contraction margin and bounded forcing. The algebraic bound alone does not certify that a fitted state or every unchosen branch is probability-valid.

Proof. The branch likelihood differs by λd, giving the lower bound on $\widehat{p}$. Subtract u/p from $(u + d)/(p + \lambda d)$ and use $u = pz^+$. The induced norm of $I - z^+\lambda$ is at most $1 + R\|\lambda\|_*$. Applying that identity at each step gives the recursion. □

Rare evidence can create a large posterior change from a small absolute error. For example, take a true unnormalized vector $(\eta, 0)$ and a perturbed vector (η, η). Their ℓ^1 difference is η, while their normalized vectors differ by one in ℓ^1. Both arise as possible branch outputs of valid probability instruments when $0 < \eta \le 1/2$, after assigning the remaining mass to another branch. There is no uniform posterior Lipschitz constant independent of a probability margin.

When a known closed convex feasible base is available, Euclidean projection of a tentative posterior back onto that base cannot increase its distance from the true posterior. This is a declared repair to the estimator, not an identity for an unmodified update. It also does not repair negative branch probabilities of arbitrary fitted operators before normalization.

36. A finite experimental route to the probability table

The following deliberately simple protocol makes a finite-sample statement possible without confusing a deterministic identity with a statistical guarantee.

Fix M finite words before estimation. For each word, reset independently to the same initial distribution, execute its prescribed action sequence, and record the indicator that all observations match. Obtain N independent trials per word. Dependencies between different words' estimates are allowed, but the N trials used for each individual mean must be independent with the declared success probability.

Proposition 35. Reset-experiment probability certificate. Let $\widehat{f}(w)$ be the sample mean for each selected word. For $0 < \delta < 1$, with probability at least $1 - \delta$, all M estimates satisfy

$$|\widehat{f}(w) - f(w)| \le \epsilon_N, \qquad \epsilon_N = \sqrt{\frac{\log(2M/\delta)}{2N}}.$$

If the selected words cover every entry of H, every G_s, and b_0, then on the same event

$$\|\widetilde{H} - H\|_2 \le r\epsilon_N, \qquad \|\widetilde{G}_s - G_s\|_2 \le r\epsilon_N, \qquad \|\widetilde{b}_0 - b_0\|_2 \le \sqrt{r}\epsilon_N.$$

These bounds feed Propositions 32 and 33 whenever their respective gap conditions hold. The empty-word probability may be set exactly to one instead of sampled.

Proof. For a centered Bernoulli variable $X - p$, write $g(t) = \log \mathbb{E} \exp(t(X - p))$. Then $g(0) = g'(0) = 0$, and $g''(t)$ is the variance of a Bernoulli under exponential tilting, at most $1/4$. Integrating twice gives $g(t) \le t^2/8$. Independence and the exponential Markov inequality give an upper-tail bound $\exp(-2N\epsilon^2)$ after optimizing $t = 4\epsilon$. The lower tail has the same bound. A union bound over the M words proves the simultaneous result. The spectral norm is bounded by the Frobenius norm, yielding the matrix and vector estimates. □

This is the Bernoulli form of Hoeffding's inequality [R11], proved here for the protocol used. It is a conservative entrywise-to-spectral bound, not a claim of optimal sample complexity. The number and lengths of chosen experiments are part of its cost.

If core tests and histories were selected from data, use a separate independent discovery sample and freeze the selected finite experiment list before these estimation trials. Conditional on that selection, the proposition applies. Alternatively, a uniform bound over a fixed larger candidate list can support data-dependent selection within that list. Reusing a post-selected table without either argument changes the claim.

A single dependent trajectory, an adaptive stopping rule, unknown resets, drifting sensors, or experiments chosen without coverage do not satisfy this proposition as stated. Those settings require their own concentration and identifiability arguments. Synthetic reset sampling in the reference example demonstrates the protocol, not its validity for every measurement source.

37. A filtering defect with a packet interpretation

The relation to the realization calculus is particularly direct before normalization. Let R be a linear summary of a hidden probability space, let $\lambda R = \mathbf{1}^\top$, and let B_s be a proposed reduced branch operator. For a true belief p, put

$$z = Rp, \qquad \delta_s = (RM_s - B_sR)p.$$

The defect δ_s is a vector in the reduced representation, not a probability of model failure. It records the failure of the unnormalized instrument to commute with the summary.

Proposition 36. Exact normalized closure defect. Suppose the true and proposed branch likelihoods

$$a = \mathbf{1}^\top M_s p > 0, \qquad b = \lambda B_s z > 0$$

are positive, and let $\widehat{z}^+ = B_s z/b$. Then

$$\boxed{R\left(\frac{M_s p}{a}\right) - \widehat{z}^+ = \frac{(I - \widehat{z}^+\lambda)\delta_s}{a}, \qquad a - b = \lambda\delta_s.}$$

If $M_s \neq 0$ and both likelihoods and normalized updates agree on every strictly positive belief, then $RM_s = B_sR$. Agreement of normalized updates alone need not force this equality.

Proof. The true unnormalized summary is $B_sz + \delta_s$, with mass $a = b + \lambda\delta_s$. Subtract the two normalized vectors and collect terms. Since a nonzero nonnegative branch has positive likelihood on every strictly positive belief, agreement permits multiplication by the common mass. This gives equality of unnormalized summaries on the interior of the probability simplex, whose linear span is the whole hidden vector space. For the last claim, a one-state model can assign different positive branch probabilities while its normalized state always remains the scalar one. □

For an orthogonal retained projector P preserving the mass row, put $Q = I - P$, $R = P$ on its image and $B_s = PM_sP$. Then

$$\delta_s = PM_sQp.$$

This is the unresolved-to-retained transfer block from Part I. Its posterior effect is not simply its norm: normalization applies the oblique map $I - \widehat{z}^+\lambda$ and divides by the true branch probability. A component proportional to $\widehat{z}^+$ changes only the branch mass and disappears from the normalized defect, while still changing likelihood-based inference over mechanisms.

A certificate should therefore retain both $\lambda\delta_s$ and the posterior defect. A posterior that looks correct can hide a wrong evidence model. This is the stochastic counterpart of the distinction between reconstruction and predictive closure.

38. A complete two-action rational example

Use two hidden states only to generate a reference process. The learner does not receive these matrices. Let

$$T_0 = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{pmatrix}, \qquad T_1 = \begin{pmatrix} 1/4 & 2/3 \\ 3/4 & 1/3 \end{pmatrix}, \qquad p_0 = \begin{pmatrix} 2/3 \\ 1/3 \end{pmatrix}.$$

The observation matrices are

$$O_0 = \operatorname{diag}(1/4, 3/4), \qquad O_1 = \operatorname{diag}(3/4, 1/4), \qquad M_{a,o} = O_o T_a.$$

The two transitions do not commute. Thus the process is genuinely action-dependent rather than one autonomous recurrence with an action label attached.

Choose tests $(\epsilon, (0,1))$ and histories $(\epsilon, (0,1))$. The observable tables give

$$H = \begin{pmatrix} 1 & 13/24 \\ 13/24 & 5/16 \end{pmatrix}, \qquad \det H = 11/576, \qquad b_0 = \begin{pmatrix} 1 \\ 13/24 \end{pmatrix}.$$

The selected coordinate q is the probability of seeing observation one after choosing action zero. In the hidden realization,

$$R = \begin{pmatrix} 1 & 1 \\ 5/8 & 3/8 \end{pmatrix}, \qquad z = Rp = (1, q)^\top.$$

The observable reconstruction $B_s = G_s H^{-1}$ yields

$$B_{0,0} = \begin{pmatrix} 1 & -1 \\ 11/32 & -1/4 \end{pmatrix}, \quad B_{0,1} = \begin{pmatrix} 0 & 1 \\ -3/32 & 3/4 \end{pmatrix},$$
$$B_{1,0} = \begin{pmatrix} 5/48 & 5/6 \\ 23/192 & 5/24 \end{pmatrix}, \quad B_{1,1} = \begin{pmatrix} 43/48 & -5/6 \\ 37/64 & -5/8 \end{pmatrix}.$$

The coefficients include negatives. Nevertheless, every operator preserves the cone $R\mathbb{R}^2_+$, because $B_s R = R M_s$. A sufficient normalized base is

$$K_0 = \{(1, q)^\top : 3/8 \le q \le 5/8\}.$$

This cone certificate uses the known generating model and is checked separately. It is not inferred merely from the finite probability table, and K_0 is not claimed to be the maximal canonical base of Proposition 29.

The likelihoods and posterior maps can now be read from the matrices. For action zero and observation one,

$$p(1 \mid z, 0) = q, \qquad q^+ = \frac{-3/32 + (3/4)q}{q} = \frac{3}{4} - \frac{3}{32q}.$$

Starting at $q_0 = 13/24$, this gives probability $13/24$ and $q_1 = 15/26$. If action one then produces observation zero, its probability is

$$\frac{5}{48} + \frac{5}{6}\frac{15}{26} = \frac{365}{624},$$

and the new predictive coordinate is

$$q_2 = \frac{23/192 + (5/24)(15/26)}{365/624} = \frac{599}{1460}.$$

The joint probability of this two-step action/observation word is $365/1152$. The reference checks compare these values and every word up to a declared finite test length against the hidden generator in exact rational arithmetic. The proposition, under its rank and coverage assumptions, is what extends the result to all word lengths.

The model's online state is not a sequence of newly measured future probabilities. It is a recursively maintained prediction initialized from learned experiment probabilities and updated after each actual action and observation. Uncertainty in those learned parameters is separate from the stochastic uncertainty in the next observation.

39. Integrated controlled stochastic realization

Proposition 37. Conditional observation-based stochastic completion. Suppose a causal finite-alphabet signal process has behavior rank at most r, and a selected $r \times r$ history/test block containing the empty test is invertible. Exact values of that block, every one-symbol shifted block, and the initial core-test vector determine a minimal realization of every future experiment probability. Its normalized state is recursively accessible after each positive-probability history, without reconstructing the hidden physical state.

The exact realization has the canonical probability cone of Proposition 29. Under any invertible atlas, transform the state, symbol operators, mass row and cone together:

$$z' = Tz, \qquad B'_s = TB_sT^{-1}, \qquad \lambda' = \lambda T^{-1}, \qquad C' = TC.$$

Every likelihood and posterior prediction is then unchanged. A complete packet decomposition preserves these predictions after full synthesis. Packetwise probability interpretation requires the extra positivity condition of Proposition 30.

For estimates obtained under Proposition 35's reset protocol, the simultaneous event there bounds the finite table perturbations. A separated finite-table spectrum yields Proposition 32's subspace certificate. A well-conditioned fixed core yields Proposition 33's operator certificate. On branches satisfying Proposition 34's likelihood margins, the resulting normalized state errors obey its recursive bound. These perturbation statements alone do not certify a learned cone or global positivity of an estimated model.

Proof. Proposition 28 reconstructs the unique minimal behavior realization, and Proposition 27 gives its conditional interpretation. Proposition 29 supplies probability validity for the exact realization. Substituting the transformed quantities cancels every $T^{-1}T$, including in the scalar normalization. Full packet synthesis is the identity, whereas Proposition 30 characterizes the additional probability-mixture condition. The perturbation conclusions are the stated chain of propositions with their separate assumptions retained. No missing gap, coverage or positivity assumption follows from matrix inversion alone. □

This theorem closes a finite controlled and stochastic sector of the foundational model. It does not claim unrestricted discovery of a nonlinear environment from a single trajectory. It does identify what a learned predictive state represents, why its operations compose, where its evidence comes from, and which defects change likelihoods or posteriors.

40. Third-block checkpoint

The model now has a single framework for exact signal archives, predictive quotients, controlled probability instruments, coordinate transport, packet decompositions, and filtering. The central mathematical object is not merely a large feature vector. It is a representation together with operations and explicit admissibility conditions under which those operations preserve meaning.

Several questions remain concrete. Finite data still require justified rank or coverage assumptions. Learning a globally invariant probability cone from partial tests is not supplied by an SVD. Continuous observations require kernels or densities with a common reference measure, and finite-alphabet proofs do not transfer by renaming their symbols. Dependence, adaptive experimentation and unknown reset

states require additional statistical arguments. Useful finite nonlinear observable dictionaries need construction and approximation theory beyond the existence of a finite-rank special case.

These are not reasons to dilute the finite conclusions. They identify the next mathematical work. In particular, the new normalized defect

$$\frac{(I - \widehat{z}^{+}\lambda)(RM_s - B_sR)p}{\mathbf{1}^{\top}M_sp}$$

provides an exact interface between representation error, packet transfer, branch evidence and posterior inference. It is the quantity a foundations-driven refinement rule would need to control, rather than an undifferentiated aggregate error.

The reference implementation is a theory laboratory, not a replacement of the production 2.3.0 runtime. The release contains exact tables, an observable-only table learner, a separate hidden-model verifier, noise calculations, reset-sampling examples, and negative controls. None is represented as Lean kernel verification. The full general proofs are the arguments in this volume.

Part IV. Certified probability domains and robust learning

A matrix learned from observations is not yet a probability model. Part III made that distinction exact: a finite predictive realization needs an admissible cone preserved by its action and observation operators. It supplied a canonical cone defined by all possible future tests, but did not supply a finite way to verify that cone for an estimated model.

This block develops a constructive finite sector of that missing step. It gives a certificate that can be checked by rational arithmetic, a search procedure that does not need hidden-state matrices, and a robust version that covers an entire declared uncertainty set. It also proves why a finite polyhedral certificate is not available for every valid finite-dimensional process. The resulting distinction is between an exact finite certificate, an uncertified candidate, and an explicitly modified approximation. Those are different mathematical outcomes.

The positive-realization and invariant-cone ingredients are classical. Jaeger's account includes the polyhedral-cone characterization and an irrational-rotation obstruction [R12]. Polyhedral invariance is also a standard subject of linear systems and optimization [R13]. The present contribution is a self-contained connection of these ingredients to Komorebi's observable learner, probability packets, confidence sets and approximation defects. The controlled rational rotation example is an adaptation of that classical obstruction, not a claim of discovering the phenomenon.

The next twelve propositions continue the numbering from the preceding volume. All have proofs below. The exact computations check stated finite instances and counterexamples. They are not a Lean formalization and do not change the production PR-LASM 2.3.0 runtime.

41. The object to be certified

Fix a real predictive space $E = \mathbb{R}^r$, a nonzero row λ, and a finite family B_s, indexed by $s = (a, o)$. The action and observation alphabets are finite and nonempty. Every declared observation branch is present, including branches represented by the zero operator. Temporal words act from right to left:

$$B_{s_1 \cdots s_k} = B_{s_k} \cdots B_{s_1}, \qquad B_\epsilon = I.$$

The mass identity is an exact constraint,

$$\lambda \sum_o B_{a,o} = \lambda \quad \text{for every action } a.$$

An initial state has $\lambda z_0 = 1$. The proposed probability of a word is $\lambda B_w z_0$. An invertible coordinate change must transform z_0, B_s, λ and the admissible cone together, as in Proposition 37.

A **probability-domain certificate** specifies a closed, pointed, generating cone $C \subset E$, with λ strictly positive on $C \setminus \{0\}$, such that

$$z_0 \in C, \qquad B_s C \subseteq C.$$

Its normalized base is $K = \{z \in C : \lambda z = 1\}$. These conditions imply nonnegative word probabilities and their per-action normalization at every depth. Positive-probability updates remain in K. Zero-probability branches have no conditional posterior and must not be assigned one by denominator flooring.

There are three separate claims. **Internal validity** says the fitted model is a stochastic model. **Data compatibility** says it agrees with the observations within a specified error budget. **Coverage of the generating process** says the assumptions and uncertainty set actually include the process that generated those observations. The first can have a deterministic certificate even when the other two remain conditional.

42. Finite tests give an outer hierarchy, not automatic closure

Define the test cones

$$C_h = \{z : \lambda B_w z \geq 0 \text{ for every } |w| \leq h\}, \qquad C_0 = \{z : \lambda z \geq 0\}.$$

Each C_h is polyhedral because it is an intersection of finitely many homogeneous half-spaces. It need not be a pointed cone or have a bounded normalized base.

Proposition 38. Test-cone hierarchy and a finite stopping certificate. The cones satisfy

$$C_{h+1} = C_h \cap \bigcap_s B_s^{-1} C_h, \qquad C_{h+1} \subseteq C_h.$$

Their intersection C_∞ is the largest cone contained in C_0 and invariant under all B_s. If an exact equality $C_{h+1} = C_h$ occurs, then $C_h = C_\infty$ and the finite inequalities certify all future word probabilities. If the

representation is observable, meaning that the rows λB_w span E^*, then C_∞ is pointed and λ is strictly positive on its nonzero elements. If reachable vectors $B_w z_0 \in C_\infty$ span E, the cone is also generating.

Proof. For a first symbol s followed by a word v, $B_{sv} = B_v B_s$. The inequalities for all words of length at most $h+1$ are therefore exactly the displayed recurrence. If $z \in C_\infty$, every test on $B_s z$ is another test on z, so C_∞ is invariant. Conversely, invariance of a cone contained in C_0 makes every successive test nonnegative, hence places it in C_∞. Equality at one finite step establishes invariance immediately and therefore gives the intersection.

If both z and $-z$ lie in C_∞, every test row annihilates z, so observability gives $z = 0$. For $z \in C_\infty$ with $\lambda z = 0$, sum word probabilities along any fixed action sequence. Mass conservation makes the sum zero. Every term is nonnegative, so every word test vanishes and again $z = 0$. Spanning by reachable cone elements gives the final assertion. □

A useful negative witness is a state admitted by the current inequalities and a specific longer word with negative proposed probability. It demonstrates that this state needs to be excluded. It does not prove that the initial state, or every reachable state, is invalid.

Finite stabilization is sufficient, not promised. Even three predictive coordinates can require a nonpolyhedral domain, as Section 51 shows. Checking a long list of trajectories is not the same as checking the invariance identity that makes every trajectory valid.

43. A finite certificate with a positive lift

Let $V = [v_1 \ \cdots \ v_m] \in \mathbb{R}^{r \times m}$ have full row rank and satisfy

$$\lambda V = \mathbf{1}_m^\top.$$

Its columns generate $C = V\mathbb{R}_+^m$ and its normalized base is $K = \operatorname{conv}\{v_1, \dots, v_m\}$. Finite generation makes the cone closed. Its normalized generators and full row rank make it pointed and generating, with a strictly positive mass functional.

Proposition 39. Finite positive-lift certificate. This cone is invariant under B_s if and only if there is an entrywise nonnegative matrix $A_s \in \mathbb{R}^{m \times m}$ such that

$$\boxed{B_s V = V A_s, \qquad A_s \geq 0.}$$

If these identities hold for every symbol and $z_0 = V\alpha_0$ for $\alpha_0 \geq 0$ with $\mathbf{1}^\top \alpha_0 = 1$, then

$$\mathbf{1}^\top \sum_o A_{a,o} = \mathbf{1}^\top, \qquad \lambda B_w z_0 = \mathbf{1}^\top A_w \alpha_0.$$

Thus the finite witness defines a nonnegative probability instrument whose image under V is the predictive model. It certifies all word probabilities without enumerating them.

Proof. Invariance places each $B_s v_j$ in the conical hull of the columns of V. Choose its nonnegative coefficients as column j of A_s. Conversely, $B_s V \alpha = V A_s \alpha \in C$ for every $\alpha \geq 0$. Multiplying the intertwining identity by λ and summing over observations gives

$$\mathbf{1}^\top \sum_o A_{a,o} = \lambda \sum_o B_{a,o} V = \lambda V = \mathbf{1}^\top.$$

Iterating $B_sV = VA_s$ gives $B_wV = VA_w$. Applying the mass row and the initial coefficients proves the word identity. Strict positivity of λ follows because a nonzero conical combination has positive total coefficient mass. □

The m nonnegative coordinates need not be physical hidden states, and m need not equal the minimal predictive dimension r. They are one finite realization witnessing validity. When V has dependent columns, the lift can be nonunique. Those redundant coordinates do not change the predictive state after synthesis.

There is a dual finite test. If a complete inequality description of the same cone is $C = \{z : Gz \geq 0\}$, then $GB_sv_j \geq 0$ for every generator j is equivalent to invariance. One may also certify $GB_s = H_sG$ with $H_s \geq 0$. The latter certificate is sufficient by direct multiplication and necessary by the polyhedral cone form of Farkas' lemma [R13]. The reference calculation uses the direct generator test, not an unreported numerical optimization certificate.

44. Robustness is a quantified invariance statement

A certificate for one fitted operator does not automatically cover neighboring operators. For uncertainty, the quantifier must range over the whole proposed model set, including correlations required by mass conservation.

Let $\mathcal{U}$ be the convex hull of finitely many **complete operator families** $\{B_s^{(j)}\}_s$. Every family must satisfy the exact per-action mass identities. The convex coefficient is shared across that family's symbols. A product of independent uncertainty sets can instead be handled by its joint vertices.

Proposition 40. Robust finite-domain certificate. For a fixed normalized generator matrix V, suppose every vertex family has witnesses

$$B_s^{(j)}V = VA_s^{(j)}, \qquad A_s^{(j)} \geq 0.$$

Then every family in $\mathcal{U}$ preserves $C = V\mathbb{R}_+^m$ and is mass conserving. If its initial state lies in K, it defines valid finite observation-tree probabilities at every horizon. This remains true when a family in $\mathcal{U}$ is selected separately at each history.

For a complete facet description $C = \{z : Gz \geq 0\}$, let $g_i^\top$ be its rows. A sufficient norm-ball certificate around $\widehat{B}_s$ is

$$\boxed{g_i^\top \widehat{B}_s v_j \geq \epsilon_s \|g_i\|_2 \|v_j\|_2 \quad \text{for all } s, i, j.}$$

It covers all perturbations with $\|B_s - \widehat{B}_s\|_2 \leq \epsilon_s$ that also satisfy the exact mass constraints.

Proof. For convex weights θ_j, take $A_s = \sum_j \theta_j A_s^{(j)}$. The equality and nonnegativity survive convex combination, as does mass conservation. At each history, every selected family maps the current cone state back into the cone. Applying the mass identity at that node proves recursive normalization, regardless of how the family was selected. For the norm ball,

$$|g_i^\top (B_s - \widehat{B}_s) v_j| \leq \epsilon_s \|g_i\|_2 \|v_j\|_2.$$

The margin makes every perturbed generator image satisfy all facets. Proposition 39, or the direct generator test, then supplies invariance. □

The switching conclusion is structural validity, not an assertion that time-varying selection represents one fixed unknown parameter. A likelihood for a fixed mechanism and a likelihood for a mechanism selected anew at each history are different statistical models.

The norm-ball inequality is sufficient and can be conservative, especially at boundary images with zero slack. Failure of this margin is not a proof that no robust certificate exists. The exact vertex test retains uncertainty correlations that a separate bound for each entry may discard.

45. Fitting inside a certified domain

Once V is fixed, certificate-constrained fitting is a convex problem. Unknowns include the entries of $\widetilde{B}_s$, A_s, $\widetilde{z}_0$ and α_0. The constraints

$$\widetilde{B}_s V = V A_s, \quad A_s \geq 0, \quad \lambda \sum_o \widetilde{B}_{a,o} = \lambda,$$
$$\widetilde{z}_0 = V \alpha_0, \quad \alpha_0 \geq 0, \quad \mathbf{1}^\top \alpha_0 = 1$$

are linear. Entrywise confidence intervals and fixed-input regression tolerances add linear inequalities. Norm balls add convex constraints. Jointly optimizing V and A_s is different: the product VA_s is then bilinear. No general convexity is claimed for learning the domain and dynamics together.

Proposition 41. Certified correction and its error budget. For fixed V, every feasible solution of the displayed constraints is an internally valid model. Suppose a center estimate $\widehat{B}_s$ and a feasible corrected estimate $\widetilde{B}_s$ satisfy

$$\|B_s^* - \widehat{B}_s\| \leq \epsilon_s, \qquad \|\widetilde{B}_s - \widehat{B}_s\| \leq \epsilon_s.$$

Then $\|\widetilde{B}_s - B_s^*\| \leq 2\epsilon_s$. The same conclusion holds for initial states in their specified norm. If the true model lies in these confidence sets and itself admits the fixed-domain certificate, the convex constraints are feasible. Feasibility alone does not establish that the true model preserves this domain.

Proof. Validity is Proposition 39. The error bound is the triangle inequality through the center estimate. Under the final assumptions, insert the true operators, true initial state and their nonnegative witnesses into the constraints. They provide a feasible point. The converse fails because existence of one valid point does not say that every point in a confidence set, or its true member, has the same invariant domain. □

This is why existential correction and robust certification must be reported separately. A corrected model can be a well-defined approximation even when no common true-model cone has been proved. The trajectory bounds below require both models to be valid on a common specified cone.

45.1 An explicit positivity repair in a simplicial cone

When V is square and invertible, put $A_s = V^{-1}\widehat{B}_s V$. Assume mass conservation has already been imposed. Let N_s be an entrywise positive reference instrument with the same per-action mass identities. The mixture

$$\widetilde{A}_s = (1-\eta)A_s + \eta N_s$$

is nonnegative provided

$$\eta \geq \max_{s,i,j:(A_s)_{ij}<0} \frac{-(A_s)_{ij}}{(N_s)_{ij} - (A_s)_{ij}}.$$

An empty maximum is zero. If every negative entry has magnitude at most d and every reference entry is at least $m_0 > 0$, the simpler sufficient choice is $\eta = d/(m_0 + d)$. These statements follow by solving each scalar inequality. Mass conservation is unchanged by convex mixing.

This is an explicit model modification, not coefficient clipping or an exact coordinate change. Relative to a true instrument on the same cone, the one-step error is at most $(1 - \eta)\epsilon + 2\eta$ in the summed base-norm metric defined next, when the original error was at most ϵ. The factor two follows because two valid instruments each distribute unit total mass. The reference example retains the repair amount as part of its approximation record.

46. A norm that measures probability mass before conditioning

Assume C is a closed, pointed, generating cone and λ is strictly positive on its nonzero elements. Its base K is compact. Define the base norm

$$\|x\|_C = \inf\{\lambda u + \lambda v : x = u - v,\ u, v \in C\}.$$

The infimum is finite because the cone is generating. Compactness of K makes this a genuine norm. Its unit ball is $\mathrm{conv}(K \cup (-K))$. In particular $\|u\|_C = \lambda u$ on C, and $|\lambda x| \leq \|x\|_C$. On the ordinary probability orthant this is the ℓ^1 norm. These are standard ordered-space constructions [R15], used here to avoid mixing Euclidean coefficient magnitude with probability mass.

Proposition 42. Instrument contraction before normalization. If $\{B_{a,o}\}_o$ is cone preserving and mass conserving for a fixed action, then

$$\boxed{\sum_o \|B_{a,o}x\|_C \leq \|x\|_C.}$$

For a polyhedral base $K = \mathrm{conv}\{v_j\}$, define the discrepancy between two instruments by

$$\epsilon_a = \sup_{z \in K} \sum_o \|(B_{a,o} - \widetilde{B}_{a,o})z\|_C.$$

The supremum equals the maximum of the same expression at the finitely many generators v_j. When both instruments are valid on C, $\epsilon_a \leq 2$.

Proof. For any $x = u - v$ with $u, v \in C$, cone preservation gives

$$\sum_o \|B_{a,o}x\|_C \leq \sum_o \lambda B_{a,o}(u + v) = \lambda u + \lambda v.$$

Take the infimum. For the generator statement, the discrepancy is convex in z. At a convex combination its value is at most the same combination of generator values, hence at most their maximum. The generators themselves belong to K, proving equality. Finally, for $z \in K$, the triangle inequality bounds the sum by the total masses of both instruments, which are one each. □

For a polyhedral certificate, the norm itself has a finite linear-program formulation:

$$\|x\|_C = \min_{a,b\geq 0}\{\mathbf{1}^\top(a+b) : V(a-b) = x\}.$$

If V is square and invertible, this reduces to $\|V^{-1}x\|_1$. The exact worked interval certificate is of this type. With dependent generators, choosing one pseudoinverse does not in general compute the base norm.

47. Whole observation laws can be stable without rare-event margins

The posterior after one rare observation can be sensitive, but that does not force the entire distribution of observation records to be equally sensitive. Work before normalization and retain the probability of each branch.

Let two models share (C, λ) and have initial states $z_0, \widetilde{z}_0 \in K$. At each time use the same policy, a specified function or randomization depending only on the observed history. Thus on any common history the action probabilities are identical in the two comparisons. This is not a comparison of two policies optimized separately for the two models.

Proposition 43. Observation-tree total-variation bound. If both instruments preserve C and conserve mass, and if every action at time t has the discrepancy in Proposition 42 bounded by ϵ_t, then the distributions $P_T, \widetilde{P}_T$ of length-T observation records under the common policy satisfy

$$\boxed{\|P_T - \widetilde{P}_T\|_{\mathrm{TV}} \leq \min\left(1, \frac{1}{2}\|z_0 - \widetilde{z}_0\|_C + \frac{1}{2}\sum_{t=0}^{T-1}\epsilon_t\right).}$$

The same bound holds for joint action and observation records when the common policy is randomized. No positive lower bound on individual branch probabilities is required.

Proof. Let $u_h, \widetilde{u}_h \in C$ be the unnormalized states on each history at depth t, including the common policy factors. Their masses sum to one in each model. Define $D_t = \sum_h \|u_h - \widetilde{u}_h\|_C$. At a history with selected action a, write

$$B_{a,o}u_h - \widetilde{B}_{a,o}\widetilde{u}_h = B_{a,o}(u_h - \widetilde{u}_h) + (B_{a,o} - \widetilde{B}_{a,o})\widetilde{u}_h.$$

Sum the first term over observations and use Proposition 42. When $\lambda\widetilde{u}_h > 0$, normalize only to apply the definition of ϵ_t to the second term. Its sum is bounded by $\epsilon_t\lambda\widetilde{u}_h$. If the mass is zero, strict positivity implies $\widetilde{u}_h = 0$. Summing histories yields $D_{t+1} \leq D_t + \epsilon_t$. Averaging over common policy probabilities gives the same inequality for randomized actions. Finally each word-probability difference is bounded by its state difference because $|\lambda x| \leq \|x\|_C$. Total variation is half the sum of those absolute differences. □

This bound separates total law error from conditional-state error. It can grow linearly with the horizon and eventually become uninformative. It is not a uniform-in-time approximation theorem. It also does not control the regret of a changed planning policy without additional decision-theoretic assumptions.

48. Conditioning still has an unavoidable cost

Let P, Q be probability distributions on a finite record space. A record event H might be a particular observed prefix. Conditional continuation laws are obtained by restriction to H and normalization.

Proposition 44. Sharp event-probability dependence of conditioning. If $\|P - Q\|_{\mathrm{TV}} \le \delta$, $p = P(H) > 0$ and $q = Q(H) > 0$, then

$$\boxed{\|P(\cdot \mid H) - Q(\cdot \mid H)\|_{\mathrm{TV}} \le \min(1, \delta/p).}$$

When $p > \delta$, positivity of q follows automatically. The factor $1/p$ cannot be removed in general.

Proof. Write the subprobability restrictions on H as u, v, of masses p, q. On the complementary event, the sum of absolute probability differences is at least $|p - q|$. Therefore

$$\|u - v\|_1 + |p - q| \le 2\delta.$$

Adding and subtracting v/p gives

$$\frac{1}{2}\|u/p - v/q\|_1 \le \frac{\|u - v\|_1 + |p - q|}{2p} \le \delta/p.$$

Also $|p - q| \le \delta$, establishing the positivity condition. For sharpness, put total mass p on an event with two outcomes. Let P assign all of it to the first, and let Q move mass $\delta \le p$ to the second, leaving the complement unchanged. The unconditional total variation is δ and the conditional distance is δ/p. □

Combining this proposition with Proposition 43 gives a conditional continuation bound when the evidence event has adequate probability. It does not give a coordinate-state error bound unless test probabilities control that coordinate norm. Validity at every branch and stable inference at every branch remain different properties.

49. Searching for a domain from observable operators

A cone certificate need not be imported from a hidden-state generator. With exact rational operators it can be searched for directly, then checked independently.

Proposition 45. Rational finite-certificate search. Suppose λ, B_s and z_0 are rational and satisfy the exact mass identities. There is an enumeration procedure that returns a valid finite certificate whenever a certificate of Proposition 39 exists with a rational normalized full-row-rank generator matrix V. The procedure is sound whenever it returns. No termination guarantee is asserted when no such rational certificate exists.

Proof. Enumerate all finite rational matrices V in an order that eventually includes every size and every bounded numerator and denominator range. Retain those of full row rank with $\lambda V = \mathbf{1}^\top$. For each retained matrix, the requirements $B_s V = V A_s$, $A_s \ge 0$, $V\alpha_0 = z_0$, $\alpha_0 \ge 0$ and $\mathbf{1}^\top \alpha_0 = 1$ are rational linear feasibility problems. A nonempty rational polyhedron has a rational feasible point, as follows from elimination of rational linear inequalities. These problems can therefore be decided by exact rational linear programming, or witnesses can be enumerated in a dovetailed rational search. Recheck every returned equality and sign exactly. Proposition 39 proves soundness. If a rational

generator certificate exists, its finite representation is eventually considered and its feasible witnesses are found. □

The general enumeration is a mathematical existence algorithm, not a practical complexity claim. A bounded candidate family is often the useful reference experiment. Failure within that family means only that the chosen family did not produce a certificate.

49.1 A certificate learned without hidden matrices

The observable tables of Section 38 determine the two-dimensional operators B_s and initial vector $(1, 13/24)^\top$. The new reference procedure receives only these fitted operators and that vector. It enumerates intervals with rational endpoints in $[0, 1]$ of reduced denominator at most eight, ordered by increasing width and then by endpoints. The coordinate restriction $[0, 1]$ comes from its declared test-probability meaning.

The first accepted interval is

$$\boxed{K = \{(1, q)^\top : 2/5 \le q \le 3/5\}.}$$

It uses

$$V = \begin{pmatrix} 1 & 1 \\ 2/5 & 3/5 \end{pmatrix}, \qquad \alpha_0 = \begin{pmatrix} 7/24 \\ 17/24 \end{pmatrix}, \qquad A_s = V^{-1} B_s V.$$

Every entry of every A_s is at least $1/64$. The full per-action column sums are exactly one. These rational identities certify all word probabilities of the fitted model, not only the words used in the fit. The certificate was not supplied to the search, and no hidden generator is imported by the standalone example.

The interval is the narrowest accepted interval **in this specified finite grid**, not a claim of the smallest invariant interval among all real endpoints. It also need not contain every predictive state associated with an arbitrary initialization of the old hidden model. It contains the declared initial state and preserves every subsequent branch of this fitted realization, which is the requirement being certified.

The positive slack gives a robust statement: any perturbation of the lifted entries by at most $1/64$, coupled to retain the exact per-action column sums, remains nonnegative. Interior perturbations of size $1/128$ are used in the worked comparison. Repeated composition is valid for all horizons, while Proposition 43 bounds the changed observation law for each fixed horizon.

50. Selection and noise do not invalidate an exact witness

A data-dependent cone proposal is not itself a statistical confidence statement. Once the matrices and witness are fixed, its exact invariance identities hold or fail independently of how it was proposed. What remains statistical is whether the chosen uncertainty set covers the generating model.

For a finite-sample interpretation, first obtain a valid uncertainty set for a fixed identifiable model class. Proposition 35 supplies one route under independent reset experiments and declared rank coverage. Proposition 33 translates those probability bounds into a fixed-core operator bound. A validity witness

can then be sought for a fitted member or robustly for every member. Only the latter identifies a common true-model domain on the coverage event.

The admissible family must enforce mass conservation exactly. A free box around unconstrained matrices generally contains nonnormalized families. Discarding those families is legitimate only because the generating process is already assumed to obey the exact mass law. Other cuts on a confidence set need their own justification.

51. Why finite predictive rank need not mean finitely many positive states

The following controlled rational example makes the boundary explicit. It is a worked adaptation of the irrational-rotation obstruction in [R12, Section 7]. Related distinctions among finite linear, positive and quantum realizations are discussed in [R14]. Only the elementary cone argument below is used.

Let $z = (t, x, y)^\top$, $\lambda = (1, 0, 0)$, $z_0 = (1, 1, 0)^\top$, and

$$R = \begin{pmatrix} 3/5 & -4/5 \\ 4/5 & 3/5 \end{pmatrix}, \qquad T = \operatorname{diag}(1, R).$$

There are two actions and two observations. A rotate action has branches

$$B_{\mathrm{r},0} = T, \qquad B_{\mathrm{r},1} = 0.$$

A probe action has branches

$$B_{\mathrm{p},0} = z_0(1/2, 1/2, 0), \qquad B_{\mathrm{p},1} = z_0(1/2, -1/2, 0).$$

The rotate outcome zero is certain. The probe has probabilities $(t + x)/2$ and $(t - x)/2$ and resets every nonzero branch to z_0 after normalization. Both actions conserve the mass t.

Proposition 46. A rational rank-three model with no polyhedral probability-domain certificate. The Lorentz cone

$$C_{\mathrm{L}} = \{(t, x, y) : t \geq \sqrt{x^2 + y^2}\}$$

is probability valid for these operators. It is the canonical all-test cone. Every closed convex invariant cone containing z_0 and admitting the nonnegative mass interpretation equals C_{L}. In particular no finite normalized generator certificate exists for this exact model. Its reachable and observable predictive dimension is three.

Proof. Rotation preserves t and $x^2 + y^2$. On C_{L} both probe effects are nonnegative and each probe image is a nonnegative multiple of z_0. Thus the displayed cone is invariant and valid.

Write the rotation angle as θ. It is not a rational multiple of 2π. Otherwise $\zeta = e^{i\theta}$ would be a root of unity, so $\zeta + \zeta^{-1} = 6/5$ would be a rational algebraic integer. Every rational algebraic integer is an integer, by the rational-root argument for a monic integer polynomial, contradicting $6/5 \notin \mathbb{Z}$. The forward powers of an irrational circle rotation are dense. Compactness supplies powers approaching the identity, so their closure contains the inverse rotation and is a closed subgroup. An infinite closed

subgroup has arbitrarily small positive angles. Integer multiples of such angles approximate every point on the circle, so that subgroup is the entire circle.

A state passing every test must pass a probe after any number k of rotations. Hence

$$t \pm e_1^\top R^k (x, y)^\top \geq 0 \quad \text{for every } k \geq 0.$$

Density gives $t \geq \sqrt{x^2 + y^2}$. The reverse inclusion was already proved, so the canonical cone is exactly C_{L}.

Any invariant cone with nonnegative mass lies inside this canonical cone, by Proposition 38. If it is closed, convex and contains z_0, it contains the orbit $T^k z_0$ and the closed convex hull of that orbit. On $t = 1$, this hull is the full unit disk. Conical scaling gives all of C_{L}. The unit disk has infinitely many extreme points, so it is not the convex hull of finitely many points.

Finally the reachable columns $z_0, Tz_0, T^2 z_0$ are independent. The mass row and probe rows before zero and one rotations span the dual space because the rotation has nonzero sine. Reachability and observability establish predictive dimension three. □

This is not a failure of finite-dimensional prediction. The exact model has only three coordinates and a simple quadratic cone. It is a failure of the narrower requirement that every valid model admit a finite *polyhedral* domain. A certificate language limited to finitely many positive generators would exclude a legitimate low-rank signal model.

No general impossibility of all finite certificates follows. This very example has a finite quadratic certificate. The obstruction concerns polyhedral certificates, not invariant cones described by other mathematics.

52. A finite approximation with an explicit price

A curved probability domain can sometimes be replaced by a finite one after changing the dynamics. That change must be quantified, not described as exact refinement.

Let P_n be the regular n-gon inscribed in the unit disk, with a vertex at $(1, 0)$ and $n \geq 3$. Replace the rotate branch by

$$T_\alpha = \operatorname{diag}(1, \alpha R), \qquad 0 \leq \alpha < 1,$$

while retaining the same probe branches. The spatial coordinates are attenuated, but total probability mass remains unchanged.

Proposition 47. Polyhedral approximation of the rotation model. If

$$\alpha \leq \cos(\pi/n),$$

then the cone over P_n is invariant under all modified operators and contains z_0. Consequently the modified model admits a finite positive lift. The original and modified models are both valid on the larger cone C_{L}, and for every common policy their length-T laws satisfy

$$\boxed{\|P_T - P_T^{(\alpha)}\|_{\mathrm{TV}} \leq \min(1, T(1-\alpha)/2).}$$

The angle-independent sufficient condition can be improved for a fixed polygon and rotation. For the rational diamond $|x| + |y| \leq t$, the present rotation is invariant after attenuation whenever $\alpha \leq 5/7$. Thus $\alpha = 2/3$ gives an entirely rational four-generator certificate.

Proof. The regular polygon contains the disk of radius $\cos(\pi/n)$ and lies in the unit disk. Every $\alpha R v$ with $v \in P_n$ has Euclidean norm at most α, hence lies in P_n under the stated condition. The probe effects remain nonnegative because P_n lies in the disk, and their normalized images are the retained vertex $(1, 0)$.

The base norm of C_{L} is

$$\|(t, x, y)\|_{C_{\mathrm{L}}} = \max(|t|, \sqrt{x^2 + y^2}).$$

Indeed the convex hull of the unit disks in the planes $t = 1$ and $t = -1$ is the cylinder with this unit ball. The rotate-branch discrepancy on a normalized state is at most $1 - \alpha$ in this norm. Probe discrepancies are zero. Proposition 43 gives the law bound. For the diamond, the induced ℓ^1 norm of R is $7/5$, so $\alpha \|R\|_1 \leq 1$ proves invariance. □

For a target attenuation loss $0 < \eta \leq 1$, choosing $\alpha = 1 - \eta$ and sufficiently many polygon vertices satisfies the certificate. The inequality $1 - \cos u \leq u^2/2$ shows that $n \geq \pi/\sqrt{2\eta}$ is sufficient, with an integer ceiling and $n \geq 3$. This is a constructive upper bound on certificate size, not a lower bound or a proof of optimality.

The distinction is exact. At $\alpha = 1$ there is no finite polyhedral certificate for this model. At $\alpha < 1$ a sufficiently fine polygon works, at the cost of a declared change in the observation law. The reference figures show that tradeoff rather than labeling the approximation an identical model.

53. Repeated proposal and verification with a confidence budget

Model development may choose another core, propose another domain or collect another independent batch after an earlier certificate is withheld. Statistical coverage must survive that selection process.

Proposition 48. Sequential confidence accounting for certified proposals. At stage n, let the proposal be measurable from previous data, and let a verification procedure return a set $\mathcal{U}_n$ for the true model in the declared identifiable coordinate system. Assume conditional coverage

$$\Pr(\text{true model} \notin \mathcal{U}_n \mid \text{previous data}) \le \delta_n, \qquad \sum_{n\ge 1} \delta_n \le \delta.$$

Then with probability at least $1-\delta$, every returned uncertainty set covers its true model. On this event, any data-selected domain with an exactly verified robust certificate for **all** members of its corresponding $\mathcal{U}_n$ is also valid for the true model, provided the true initial state is covered by the certified base. The conclusion remains valid at any stopping time chosen from the recorded results.

Proof. Take expectations of the conditional failure bounds, then apply the countable union bound. Off the union of failure events, all coverages hold simultaneously. Robust invariance over an entire set then applies to its true member by deterministic implication, independently of how its domain was chosen. A stopped stage is one of the already covered stages. □

For example, $\delta_n = \delta 2^{-n}$ spends a finite total confidence budget. Fresh reset trials for a proposal fixed by past data can use Proposition 35 conditionally, with this stage budget. An exact witness needs no statistical correction for how many candidate cones were tried. The coverage statement does need correction for repeatedly constructing confidence sets or inspecting post-selected data.

The proposition does not infer a global rank bound, validate an unknown reset operation, or transform dependent observations into independent trials. Those are assumptions of the coverage procedure. A known coordinate transformation can transport a confidence set exactly. An estimated transformation requires the subspace and conditioning controls already separated in Part III.

54. Integrated certificate-carrying realization

Proposition 49. Conditional learning with an all-horizon probability certificate. Suppose the observable identification assumptions of Proposition 37 hold for a declared finite-rank controlled process. Suppose a justified confidence procedure produces an operator and initial-state set containing the true realization on an event of probability at least $1-\delta$. Assume the per-action mass constraints are exact.

If a fitted realization admits a checked finite witness of Proposition 39, it is an internally valid stochastic model at every horizon. If the witness is robust over the whole confidence set and covers its initial states, then on the coverage event the same probability domain is valid for both the fitted and true models. When their summed base-norm instrument discrepancies are at most ϵ_t and their initial discrepancy is at most e_0, every common policy has the bound

$$\|P_T - \widehat{P}_T\|_{\mathrm{TV}} \le \min\left(1, \frac{e_0 + \sum_{t<T} \epsilon_t}{2}\right).$$

Conditioning on a positive-probability event then obeys Proposition 44. Exact coordinate transport preserves the certificate. A signed packet decomposition preserves it after complete synthesis but does not turn individual packets into probability states without Proposition 30's extra positivity requirement.

Proof. Observable identification supplies the specified realization and the confidence procedure supplies the stated event, not an inferred event of its own. Proposition 39 supplies internal validity for the fitted model. Proposition 40 extends the same domain to every member of the uncertainty set, hence to its true member on that event. Propositions 42 and 43 give the total-variation bound, and Proposition 44 gives the conditional bound. For a coordinate change T, use $V' = TV$, $B'_s = TB_sT^{-1}$ and $\lambda' = \lambda T^{-1}$. Then $B'_sV' = V'A_s$ and $\lambda'V' = \mathbf{1}^\top$ with the unchanged nonnegative witnesses. Full packet synthesis is the identity. □

This closes a concrete foundational gap. A learner can now return not just a matrix family but a verifiable domain in which that family has probability meaning. It can attach a robust uncertainty statement when one is justified, or return an internally valid approximation with its correction budget explicitly recorded.

It does not turn a successful fit into unrestricted knowledge of the world. The model class, observation protocol, rank coverage, exact mass law, domain witness and approximation budget remain visible parts of the result. Failure of a finite search is a withheld certificate, not a proof of invalidity. The nonpolyhedral example explains why this distinction is necessary even for very small exact models.

55. Foundation status after this block

The cumulative theory now contains 49 propositions. This block adds a finite route from observable operators to probability-domain certificates, robust invariance, certified correction, pre-normalization stability of full observation laws, and a precise boundary for finite positive lifts. It also supplies an observable-only interval search and a rational curved-domain counterexample with a quantified polyhedral approximation.

These are not twelve claims of historical novelty. The positive-realization and rotation ingredients have explicit antecedents [R12, R14]. Convex invariance and ordered-space contraction are established mathematics [R13, R15]. The value of the block is the connected set of requirements, constructive witnesses, exact counterexamples and uncertainty accounting within the large signal model.

The next open tasks are no longer simply "make the probabilities valid." They include efficient synthesis of useful nonpolyhedral domains, informative confidence regions in poorly conditioned predictive coordinates, dependent-data and adaptive-intervention learning, continuous observation measures, and nonlinear representations whose discovery and approximation costs are controlled. A general practical solver for all such domains is not supplied here.

The production runtime remains unchanged. The new code is an exact reference checker, a bounded rational domain search and finite verification experiments. General validity comes from the stated proof and its checked witness, not from a large number of sampled trajectories. The release ledger separates unchanged predecessor source, new results, finite arithmetic evidence, expected rejections and unexecuted work.

References and source lineage

[**K1**] Komorebi P. *Projector-Resolved Large Algebraic Signal Model*. Historical 38-page capstone preserved in `Komorebi_PR_LASM_2.3.0_Source.zip`, `paper/Projector_Resolved_Large_Algebraic_Signal_Model.tex`. In particular, Sections 2–7 and the categorical-atlas distinction. Historical experimental claims are not used here.

[**K2**] Komorebi P. *Projector-Resolved Seed Calculus VI*, corrected presentation in `Paper_VI_With_Visual_Appendix.pdf`, Section 8, Theorem 8.4 and the adjacent coarsening criterion and one-packet boundary. This supplement restates and proves the finite result it uses. It does not claim a fresh Lean audit of that paper.

[**K3**] Komorebi P. *PR-LASM 2.3.0 Release Audit*, 9 October 2026, and the accompanying source. The source inspected includes `exact_atlas.py`, `lasm_packetizer.py`, `lasm_model.py`, `docs/RELEASE_GUIDE.md` and `docs/claim-boundary.md`.

[**R1**] Ronald R. Coifman and Mladen Victor Wickerhauser. "Entropy-based algorithms for best basis selection." *IEEE Transactions on Information Theory* 38(2), 713–718, 1992. DOI: 10.1109/18.119732. Source checked: the authors' Washington University research record. The finite additive tree optimization is classical background.

[**R2**] MIT OpenCourseWare. *6.241J Dynamic Systems and Control*, Chapter 24, "Observability," Spring 2011, especially Section 24.5 on the largest invariant subspace contained in the observation kernel. Source checked: the MIT course notes, PDF page 6. Used to identify the classical control-theoretic antecedent of Proposition 8.

[**R3**] Michael Woodward, Yifeng Tian, Arvind Mohan, Yen Ting Lin, Christoph Hader, Hermann Fasel, Misha Chertkov and Daniel Livescu. *Data-Driven Mori-Zwanzig: Approaching a Reduced Order Model for Hypersonic Boundary Layer Transition*, arXiv:2301.07203, version 1, 2023, introduction and description of resolved, memory and unresolved contributions. Cited for the established projection-memory context, not as evidence for PR-LASM's performance or for any application discussed in that paper. Proposition 10 is proved directly by finite linear recurrence expansion here.

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[**R8**] Stephen Boyd, Laurent El Ghaoui, Eric Feron and Venkataramanan Balakrishnan. *Linear Matrix Inequalities in System and Control Theory*. SIAM Studies in Applied Mathematics 15, 1994. The authors' Stanford publication page identifies the book and provides the primary text. Used to identify the classical common quadratic Lyapunov and matrix-inequality setting. The discrete contraction implication used in Section 19 is proved directly there.

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[**R11**] Wassily Hoeffding. "Probability Inequalities for Sums of Bounded Random Variables." *Journal of the American Statistical Association* 58(301), 13–30, 1963. DOI: 10.1080/01621459.1963.10500830. Publisher record checked. Proposition 35 supplies a self-contained Bernoulli moment-generating-function proof for the protocol it uses. Source: `https://www.tandfonline.com/doi/abs/10.1080/01621459.1963.10500830`.

[**R7a**] Daniel Hsu, Sham M. Kakade and Tong Zhang. "A Spectral Algorithm for Learning Hidden Markov Models." *COLT*, 2009. Original conference manuscript inspected at the official Learning Theory proceedings site, particularly Sections 2.3–3.1 on rank, sampling and observable representations. The later journal version is [R7]. No statistical guarantee from either version is imported without its assumptions. Source: `https://www.learningtheory.org/colt2009/papers/011.pdf`.

[**R12**] Herbert Jaeger. "Observable operator models for discrete stochastic time series." *Neural Computation* 12(6), 1371–1398, 2000. Author-hosted manuscript with the August 2017 erratum. Section 7 gives the polyhedral-cone connection to finite positive realizations and an irrational-rotation obstruction. The controlled rational example in Section 51 is an independently worked adaptation of that established obstruction, not a claim of discovering it. The corrected manuscript and its geometric argument were inspected. Source: `https://www.ai.rug.nl/minds/uploads/oom_neco00.pdf`.

[**R13**] Zoltán Horváth, Yunfei Song and Tamás Terlaky. *A Novel Unified Approach to Invariance Conditions for a Linear Dynamical System.* arXiv:1405.5167, version 2, 5 July 2016. Primary author manuscript, particularly the polyhedral-invariance discussion and use of Farkas' lemma. Used for the classical invariant-cone and optimization setting. The generator and robust-vertex statements used here are proved directly. Source: `https://arxiv.org/html/1405.5167v2`.

[**R14**] Marco Fanizza, Josep Lumbreras and Andreas Winter. *Quantum theory in finite dimension cannot explain every general process with finite memory.* arXiv:2209.11225, version 2, 5 May 2023. Primary manuscript, especially its discussion of finite linear realizations, positive realizations and polyhedral cones. Cited for the distinction between these representation classes. No quantum implementation or quantum advantage is claimed for PR-LASM. The present block uses a real three-dimensional cone construction proved in Section 51. Source: `https://arxiv.org/html/2209.11225v2`.

[**R15**] Anna Jenčová. *Base norms and discrimination of generalized quantum channels.* arXiv:1308.4030, version 1, 19 August 2013. Primary manuscript, Section 1.2, printed page 5, on finite-dimensional proper cones, normalized bases and base norms. That ordered-vector-space definition is used in Section 46. The classical instrument contraction and observation-tree bound needed here are derived in the text, without importing quantum-channel assumptions. Source: `https://arxiv.org/pdf/1308.4030`.